

Fractional Brownian Motion for Rough Volatility Modelling

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Contents

0.1	Introduction	4
1	Fractional Brownian Motion	5
1.1	Preliminaries	5
1.1.1	Fractional Calculus	5
1.1.2	Hölder Continuity	7
1.2	Fractional Brownian Motion	8
1.2.1	Definition & Motivation	8
1.2.2	Correlation & Long-Range Dependency	9
1.2.3	Semi-Martingale Property	11
1.2.4	Markov Property	12
1.3	Mandelbrot-Vann Ness representation	12
1.4	The Ornstein-Uhlenbeck Representation	15
1.4.1	Ornstein-Uhlenbeck Process	15
1.4.2	Multivariate Ornstein-Uhlenbeck process	16
1.4.3	Infinite Dimensional OU proccess & the Markov Property	18
1.4.4	Derivation of Muravlev Representation	19
1.5	Simulating fBm	21
1.5.1	Cholesky Method	21
1.6	Markovian Lifting	22
1.6.1	Fractional Kernel Approximation for $H < \frac{1}{2}$	22
1.6.2	Lifted fBm for $H < \frac{1}{2}$	24
1.6.3	Fractional Kernel Approximation for $H > \frac{1}{2}$	25
1.6.4	Lifted fBm for $H > \frac{1}{2}$	27
2	Rough Volatility Modelling	31
2.1	Black-Scholes & Implied Volatility	32
2.2	The Heston Model	32
2.3	Volatility is Rough	34
2.4	The Rough Heston Model	34
2.5	The Lifted Heston Model	35
2.5.1	Markovian Approximation of Rough Heston	35
2.5.2	Convergence to Rough Heston Model	38
3	Conclusion & Future Developments	41

A	Simulating the OU Process	43
B	Lifted Heston Characteristic Function	44

Notation

$\mathbb{N}$	natural numbers $\{1, 2, 3, \dots\}$
$\mathbb{R}^+$	non-negative reals $[0, \infty)$
$X \stackrel{d}{=} Y$	equality in distribution of random variables
I_a^α	left fractional integral of order α over $[a, b]$
D_a^α	left fractional derivative of order α over $[a, b]$
B_t^H	Fractional Brownian motion with hurst parameter H
I_m	$m \times m$ identity matrix
$\mathbf{A}^T$	transpose of the matrix $\mathbf{A}$
$diag(a_1, \dots, a_m)$	diagonal matrix with entries $a_1, \dots, a_m$
$x_n \rightarrow a$	x_n The series converges to a as $n \rightarrow \infty$
$x_n \downarrow a$	x_n converges to a from above
$x_n \uparrow a$	x_n converges to a from below
$x \wedge y$	minimum of x, y
$x \vee y$	maximum of x, y
B_t^H	fractional Brownian motion with Hurst parameter H at time t
Z_t^β	standard Ornstein-Uhlenbeck process with mean reversion β at time t

0.1 Introduction

Fractional Brownian motion (fBm_H) has emerged as an indispensable tool in modern quantitative finance, particularly in the characterization of asset price volatility. Despite its widespread adoption by both theoretical mathematicians and financial practitioners, there remains a pronounced pedagogical gap: accessible, self-contained treatments of fBm_H tailored to an advanced undergraduate level are notably scarce. Existing literature (see, e.g., [15], [8], [13], [11]) typically falls into one of two categories. They are either couched in a degree of abstraction far exceeding the needs of a practitioner, or they introduce key integral representations axiomatically, without providing the underlying constructive intuition. This project aims to bridge this gap by providing a rigorous yet accessible treatment of fBm_H , developing its integral representations naturally from core principles.

The mathematical centerpiece of this work is the theory underlying the infinite-dimensional Ornstein-Uhlenbeck (OU) representation for fractional Brownian motion. We leverage this framework to simulate fBm_H paths. While the general OU-superposition representation is known for all H (see [16]), and alternative numerical schemes exist for $H > 1/2$ (see [3]), the specific geometric quadrature proposed for the Lifted Heston model in [7] has, to our knowledge, only been analysed in the sub-diffusive ($H < \frac{1}{2}$) case. Our contribution is to apply this specific discretization to the super-diffusive ($H > \frac{1}{2}$) regime and empirically verify its convergence.

To demonstrate the practical relevance of this theory, Chapter 2 illustrates how fBm_H has revolutionized contemporary volatility modelling. We introduce the classical Heston model alongside its rough counterpart, the rough Heston model. We then focus on an active area of contemporary research: the *lifted Heston model*. This framework utilizes the exact infinite-dimensional OU representation developed in Chapter 1 to construct a tractable, finite-dimensional Markovian approximation to the otherwise non-Markovian rough Heston model.

The remainder of this report is structured as follows. Chapter 1 develops the foundational theory of fBm_H from first principles, establishing its core properties before naturally deriving its integral representations - beginning with the classical Mandelbrot Van-Ness representation and culminating in the infinite-dimensional OU representation. Chapter 2 applies this theoretical machinery to financial volatility modelling, drawing on contemporary literature to provide an exposition of the lifted Heston model. We explicitly note that a rigorous treatment of stochastic differential equations directly driven by fBm_H falls outside the scope of this text; instead, our focus remains centered on the mechanics and validation of the Markovian lifting approximation technique.

Chapter 1

Fractional Brownian Motion

1.1 Preliminaries

1.1.1 Fractional Calculus

Before going into the definition & representations of fBm, it will be heuristically important for us to first understand fractional calculus. The idea behind fractional calculus is to give a sensible definition for what it means to have an integral or derivative of fractional order. Following [10] we begin by looking at the Cauchy formula for iterated integration.

Theorem 1.1.1 (Cauchy n -Fold Integral Formula). *Given any $a, x \in \mathbb{R}$ and an integrable function f , the following formula holds for all $n \in \mathbb{N}$*

$$(I_a^n f)(x) := \int_a^x \int_a^{x_1} \cdots \int_a^{x_{n-1}} f(x_{n-1}) dx_{n-1} \cdots dx_1 dx = \frac{1}{(n-1)!} \int_a^x (x-t)^{n-1} f(t) dt$$

Proof. We prove this by induction on n . The base case $n = 1$ is trivial. Suppose that for $n - 1$ we have

$$(I_a^{n-1} f)(x) = \frac{1}{(n-2)!} \int_a^x (x-t)^{n-2} f(t) dt.$$

Then by definition we have

$$\begin{aligned}
(I_a^n f)(x) &= \int_a^x (I_a^{n-1} f)(s) ds \\
&= \frac{1}{(n-2)!} \int_a^x \int_a^s (s-t)^{n-2} f(t) dt ds \\
&= \frac{1}{(n-2)!} \int_a^x f(t) \int_t^x (s-t)^{n-2} ds dt \quad (\text{By Fubini's theorem}) \\
&= \frac{1}{(n-2)!} \int_a^x f(t) \cdot \frac{(x-t)^{n-1}}{n-1} dt \\
&= \frac{1}{(n-1)!} \int_a^x (x-t)^{n-1} f(t) dt.
\end{aligned}$$

□

Now keeping in mind that the gamma function ¹ satisfies $\Gamma(n) = (n-1)!$ for all non-negative integers, it seems that using this would give natural extension of our integral formula to the positive reals [10].

Definition 1.1.1 (Left Hand Riemann-Liouville Fractional Integral). Let $I = [a, b]$ be a finite interval on the real axis $\mathbb{R}$. The left-side fractional integral $I^\alpha f$ of order² $\alpha \in \mathbb{R}^+$ is defined for $x > a$ by

$$(I_a^\alpha f)(x) = \frac{1}{\Gamma(\alpha)} \int_a^x \frac{f(t)}{(x-t)^{1-\alpha}} dt$$

Now by noting that we want derivatives and integrals to be “inverses” of one another, we expect any reasonable definition of the fractional derivative to satisfy $(D^\alpha(I^\beta f))(x) = (I^{b-\alpha} f)(x)$.

Definition 1.1.2 (Left Hand Riemann-Liouville Fractional Derivative). The left-side fractional derivative $D^\alpha f$ of order $\alpha \in \mathbb{R}$ on $[a, b]$ is defined by

$$\begin{aligned}
(D_a^\alpha f)(x) &= \left(\frac{d}{dx} \right)^n (I_a^{n-\alpha} f)(x) \\
&= \frac{1}{\Gamma(n-\alpha)} \left(\frac{d}{dx} \right)^n \int_a^x \frac{f(t) dt}{(x-t)^{\alpha-n+1}}.
\end{aligned}$$

Where we take any natural $n > \alpha$, thus the standard is to use $n = [\alpha] + 1$ which is the smallest integer larger than α .

¹ $\Gamma(x) := \int_0^\infty t^{x-1} e^{-t} dt$

²Integrals of complex order can also be defined but we choose to omit that here as it is not relevant to our work, although is interesting in its own right.

1.1.2 Hölder Continuity

If we want to look at continuous time process it will be important that we have a way of measuring how “rough” the sample paths are. This motivates the idea of Hölder continuity.

Definition 1.1.3 (Hölder Continuity [5]). Let $0 < \gamma \leq 1$. A function $f : [0, T] \rightarrow \mathbb{R}$ is called uniformly Hölder continuous with exponent $\gamma > 0$ if there exists a constant K such that

$$|f(t) - f(s)| \leq K|t - s|^\gamma \quad \text{for all } s, t \in [0, T].$$

We say f is Hölder continuous with exponent $\gamma > 0$ at the point s if there exists a constant K such that

$$|f(t) - f(s)| \leq K|t - s|^\gamma \quad \text{for all } t \in [0, T].$$

From a heuristic point of view, γ tells us how much the functions oscillate over a small interval. Using sufficiently small time intervals $|t - s| = \delta > 0$ it follows that δ^γ increases as γ decreases. With $\gamma = 1$ giving us a Lipschitz function, which is known to be differentiable almost everywhere³, as $\gamma \rightarrow 0$ the bound on oscillation becomes independent of $|t - s|$, meaning the function need not be continuous at all.

We will see that the sample paths of fBm are Hölder continuous with exponent $\gamma < H$, where H is the Hurst parameter, making this a natural tool for measuring path roughness. A good general theorem which we will later use to prove Hölder continuity of fBm paths is this important theorem of Kolmogorov of which we do not include a proof of here.

Theorem 1.1.2 (Kolmogorov-Chentsov continuity theorem [15]). *Assume that for a stochastic process $(X_t)_{t \geq 0}$ there exist $K > 0, p > 0, \beta > 0$ such that for all $t \geq 0, s \geq 0$*

$$\mathbb{E}[|X_t - X_s|^p] \leq K|t - s|^{1+\beta}.$$

Then the process X has a continuous modification⁴. Moreover, for any $\gamma \in (0, \beta/p)$ and $T > 0$ the modified process $\tilde{X}$ is γ -Hölder continuous on $[0, T]$.

Heuristically this theorem just states the implications between the Hölder regularity of the p -th moment and the random itself.

We lastly show that fractional integration increases Hölder continuity by order of integral to motivate time representation of fBm in 1.3.

Theorem 1.1.3 (Integration increases Hölder regularity). *Suppose $0 < \gamma \leq 1$ and $\alpha \in \mathbb{R}$ satisfy $0 < \gamma + \alpha \leq 1$. Further suppose that $f : [a, b] \rightarrow \mathbb{R}$ is γ -Hölder continuous, then $I^\alpha f$ is $(\gamma + \alpha)$ -Hölder continuous.*

This proof uses standard estimates which we choose to omit (see [14]).

³This follows from Rademacher's theorem, which states that any Lipschitz function is differentiable almost everywhere (see [5]).

⁴i.e. a process $\{\tilde{X}_t, t \geq 0\}$ such that $\tilde{X} \in C[0, \infty)$ and for all $t \geq 0$ $\mathbb{P}(X_t = \tilde{X}_t) = 1$.

1.2 Fractional Brownian Motion

The fruits of fractional calculus will now pay off as it is an essential tool used in representing fBm as we will now come to see.

1.2.1 Definition & Motivation

The assumptions of independent, stationary increments along with continuous paths give Brownian motion. Now suppose we remove the dependent increments property and want to keep self-similarity. For this we will now define a new process - fBm - for which we will retain the assumptions of stationary increments and continuous paths, but will allow the increments to be dependent. The process B_t^H we construct depends on a parameter $H \in (0, 1)$ called the Hurst index. It measures the correlation of the increments. We will also assume that the process is self-similar⁵

- If $a > 0$, then the distribution of

$$X_t := a^{-H} B_{at}^H$$

is the same as that of B_t^H . In particular, B_t^H has the same distribution as $t^H B_1^H$.

Our final assumption is that the process is a centred Gaussian process.

- For every $t_1, \dots, t_k$, the random vector $(B_{t_1}^H, \dots, B_{t_k}^H)$ has a joint normal distribution with mean zero.

We normalize so that $\text{Var}(B_1^H) = \mathbb{E}[(B_1^H)^2] = 1$ and hence

$$\text{Var}(B_t^H) = \text{Var}(t^H B_1^H) = t^{2H} \text{Var}(B_1^H) = t^{2H}.$$

Thus since we would want increments to be stationary⁶ it seems reasonable to have

$$\begin{aligned} \mathbb{E}[(B_t^H - B_s^H)^2] &= |t - s|^{2H} \\ \mathbb{E}[(B_t^H)^2] + \mathbb{E}[(B_s^H)^2] - 2\mathbb{E}[B_t^H B_s^H] &= |t - s|^{2H}. \end{aligned}$$

This motivates the covariance structure upon which we will now define fBm.

Definition 1.2.1 (Fractional Brownian Motion). If $H \in (0, 1)$, the fractional Brownian motion with Hurst parameter H (fBm _{H}) is the centred (mean zero) Gaussian process B_t^H with continuous paths such that for all $s, t \geq 0$

$$\text{Cov}(B_s^H, B_t^H) = \mathbb{E}[B_s^H B_t^H] = \frac{1}{2} [s^{2H} + t^{2H} - |s - t|^{2H}].$$

If $H = 1/2$, then fractional Brownian motion is the same as usual Brownian motion.

⁵Any self-similar process must be of this form, since if $X_{at} = \lambda X_t$ then it follows that $X_{at} X_{a't} = \lambda \lambda' X_t$ which clearly gives us a multiplicative functional equation which can easily be shown to only have solutions satisfying a power law.

⁶A stochastic process is stationary if it's distribution is the same at all times, thus for a centred Gaussian process it suffices that the variance is constant.

We will prove in then next section 1.3 using an integral representation that such a process actually exists, but for now we assume it does exist. We next state some theorems to build up some intuition on the behaviour of fBm.

Theorem 1.2.1 (Covariance Between Increments). *If $0 \leq s < t < \infty$, note that*

$$\begin{aligned}\mathbb{E} [B_s^H (B_t^H - B_s^H)] &= \mathbb{E} [B_s^H B_t^H] - \mathbb{E} [(B_s^H)^2] \\ &= \frac{1}{2} [t^{2H} - s^{2H} - (t-s)^{2H}]\end{aligned}$$

This covariance structure intuitively can be explained by the following idea on how rough the paths are for each H . Namely since $B_{t+\delta}^H - B_t^H \sim N(0, \delta^{2H})$, we can write roughly

$$|B_{t+\delta}^H - B_t^H| \approx \delta^H$$

In other words, the Hölder exponent of fBm_H is given by the Hurst index (just under) H .

If $H > 1/2$, the paths are “smoother” than Brownian paths and if $H < 1/2$, the paths are “rougher”. More formally we have the following theorem.

Theorem 1.2.2 (Hölder Continuity of fBm_H). *If $(B_t^H)_{t \geq 0}$ is a fBm_H process then it is γ -Hölder continuous for all $\gamma \in (0, H)$.*

Proof. We use the Kolmogorov–Chentsov continuity theorem 1.1.2 as follows. Fix $t > s$. Since B^H is Gaussian and centered, we have that

$$B_t^H - B_s^H \stackrel{d}{=} \sqrt{\mathbb{E} [|B_t^H - B_s^H|^2]} Z,$$

where $Z \sim N(0, 1)$. So by recalling that $\mathbb{E} [|B_t^H - B_s^H|^2] = |t - s|^{2H}$ then we have

$$B_t^H - B_s^H = |t - s|^H Z$$

So given any $p \in \mathbb{N}$ we have that

$$\mathbb{E} [|B_t - B_s|^p] = \mathbb{E} [|Z|^p] |t - s|^{pH}$$

We deduce from the Kolmogorov–Chentsov lemma (Theorem 1.1.2) by noting that $K_p = \mathbb{E} [|Z|^p]$ is constant for each p and by taking $\beta = pH - 1$ the result follows for all $0 < \gamma < \frac{\beta}{p} = H - \frac{1}{p}$. Hence by taking p so that $pH > 1$ we have that B_t^H has Hölder regularity γ for all $\gamma \in (0, H)$. $\square$

1.2.2 Correlation & Long-Range Dependency

A fBm_H is clearly not a stationary process (variance of increments changes over time). However it was baked into the way we defined fBm_H in Section 1.2.1 that the increment process is stationary which allows us to talk about the autocorrelation between increments of arbitrary times. In this subsection we will

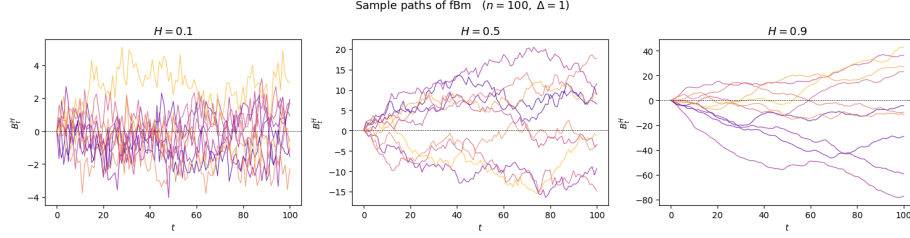


Figure 1.1: fBm sample paths

show some properties of the dependence of the increments of fBm, and also exhibit when the fBm has long-range dependence (long-memory).

We will start by looking at the autocorrelation between increments. Consider the following discrete-time Gaussian process $X_n = B_n^H - B_{n-1}^H, n \in \mathbb{N}$, known as a fractional Gaussian noise (fGn), a generalisation of a traditional white noise corresponding to the case where $H = \frac{1}{2}$. Here X_k is a sequence of $N(0, 1)$ random variables that are not necessarily independent of one another.

If we consider $\rho_n = \mathbb{E}(X_{k+n}X_k)$, we can observe that it depends only on n , and not k . We have that

$$\begin{aligned} \rho_n &= E(X_{k+n}X_k) = E((B_{k+n}^H - B_{k+n-1}^H)(B_k^H - B_{k-1}^H)) \\ &= \frac{1}{2} [(n+1)^{2H} - n^{2H} - (n^{2H} - (n-1)^{2H})]. \end{aligned}$$

We now pose the question as to what this autocorrelation structure means for fBm_H over long periods of time. Heuristically long range dependence (LRD) is just the idea that a stochastic process' current value can have long term effects on the future value, or in other words the increments between two distant values may have sufficiently large correlation. Many different definitions for LRD are used in the literature, we will use the following to define LRD:

Definition 1.2.2 (Long Range Dependence). Suppose $(X_t)_{t \geq 0}$ is a stationary process and let $\rho_n := \text{Cov}(X_t, X_{t+n})$ which is independent of t since the process is stationary. We say that X has the long range dependence property if

$$\sum_{n=1}^{\infty} \rho_n = \infty.$$

The motivation for this definition is rooted in statistical inference where this sum is important when computing statistics such as the sample mean or variance.

Theorem 1.2.3 (Long range dependence of fBm_H). *fBm_H has long range dependence for $H \in (\frac{1}{2}, 1)$ and short range dependence for $H \in (0, \frac{1}{2})$.*

Proof. We prove this by first noting the Taylor expansion of $(1+x)^{2H}$ for small $x = \frac{1}{n}$ allowing us to rewrite ρ_n as follows.

$$\begin{aligned}\rho_n &= \frac{1}{2}n^{2H} \left[\left(1 + \frac{1}{n}\right)^{2H} + \left(1 - \frac{1}{n}\right)^{2H} - 1 - 1 \right] \\ &= \frac{1}{2}n^{2H} \left[\left(1 + 2H\frac{1}{n} + \frac{1}{2} \cdot 2H(2H-1)\frac{1}{n^2}\right) \right. \\ &\quad \left. + \left(1 - 2H\frac{1}{n} + \frac{1}{2} \cdot 2H(2H-1)\frac{1}{n^2}\right) - 1 - 1 + O\left(\frac{1}{n^3}\right) \right] \\ &= H(2H-1)n^{2H-2} + O(n^{2H-3})\end{aligned}$$

Now first notice that since $2H-3 < -1$ for all $H \in (0,1)$ it does not effect the convergence of our sum⁷. And for $H \in (\frac{1}{2}, 1)$ it follows that $2H-2 > -1$ so

$$\sum_{n=0}^{\infty} \rho_n = \infty \quad \text{for } H > \frac{1}{2}$$

and similarly this series converges for $H < \frac{1}{2}$. We note also that the $H = \frac{1}{2}$ case is just our classical Brownian motion which has $\rho_n = 0$ so we have short range dependence too. $\square$

1.2.3 Semi-Martingale Property

In this section we very briefly demonstrate that fBm_H is not integrable in the sense of the classic stochastic Ito (or Stratonovich integral) integral. We leave out the details so as to keep the main idea in mind going forward to the mathematical core in the next section.

Definition 1.2.3 (Semimartingale). A process $X = X_t, t \geq 0$ with associated filtration $\mathcal{F}_t$ is called a semimartingale if it admits the following representation:

$$X_t = X_0 + M_t + Y_t,$$

where X_0 is $\mathcal{F}_0$ -measurable, M is a $\mathcal{F}_t$ -local martingale with $M_0 = 0$, and Y_t is a process of locally bounded variation.

Lemma 1.2.4. A semimartingale X_t has bounded quadratic variation.

With this at hand, we can address the non-semimartingale property of fBm .

Theorem 1.2.5. Let B_t^H be a fBm with Hurst index $H \in (0,1)$. Then, for $H \neq \frac{1}{2}$, B_t^H is not a semimartingale.

A very brief sketch of the proof of this theorem is by demonstrating that fBm_H has unbounded quadratic variation.

⁷This follows from the p -test for convergent series which states that $\sum n^p$ converges if and only if $p < -1$ and noting that adding a convergent term to the inside of our series doesn't effect it's convergence

1.2.4 Markov Property

A stochastic process is said to be a Markov process if its values at time s encapsulates all relevant future information.

Definition 1.2.4 (Markov property). A stochastic process $(X_t)_{t \geq 0}$ is said to be a Markov process if

$$X_t | \mathcal{F}_s \stackrel{d}{=} X_t | X_s$$

where $\mathcal{F}_s = \sigma(X_u : u \leq s)$ is the sigma algebra generated by X up to time s , also known as the history of X up to time s .

One of the key problems that makes fBm_H difficult to work with, especially when it comes to financial modelling, is the fact that it is non-markovian. In section 1.4.4 we will demonstrate a slick workaround for this non-markovian nature, thus to motivate the necessity of the representation 1.4.4 we prove here this non-markovian property. The following proof is an original and highly simplified version of that given in [13].

Theorem 1.2.6 (fBm_H is non-Markovian [13]). *For $H \neq \frac{1}{2}$ fBm_H is not a markov process*

Proof. Suppose, for contradiction, that B_t^H is Markovian. The following identity for the Gaussian Markovian random variable holds (see [13]):

$$E [B_s^H B_u^H] E [(B_t^H)^2] = E [B_s^H B_t^H] E [B_t^H B_u^H].$$

for any $0 \leq s \leq t \leq u$. Hence for $s = 1, t = 2, u = 3$ we get

$$\begin{aligned} E [B_1^H B_3^H] E [(B_2^H)^2] &= E [B_1^H B_2^H] E [B_2^H B_3^H] \\ \implies \frac{1}{2} [1^{2H} + 3^{2H} - 2^{2H}] \times 2^{2H} &= \frac{1}{2} [1^{2H} + 2^{2H} - 1^{2H}] \times \frac{1}{2} [2^{2H} + 3^{2H} - 1^{2H}] \\ \implies 2^{2H} - 3^{2H-1} - 1 &= 0 \end{aligned}$$

Now we can show⁸ that $H = \frac{1}{2}$ is the unique solution to the above equation for $H \in (0, 1)$. Hence we have a contradiction of our Markovian assumption for $H \neq \frac{1}{2}$. The $H = \frac{1}{2}$ case is simply standard Brownian motion which is known to be Markovian. $\square$

1.3 Mandelbrot-Vann Ness representation

So we have seen that fBm_H is a Gaussian process that is (just under) H -Hölder continuous. Moreover we have seen in section 1.1.1 that fractional integration increases the Hölder parameter of our integrand by the order of integration. Thus it seems natural to try to directly represent fBm_H via an integral in this

⁸To see this is the unique solution we have merely used graphical methods.

way. We start by noting from [5] that we can represent a standard Brownian motion $(B_t)_{t \geq 0}$ by the stochastic integral

$$B_t = \int_0^t dB_s.$$

This heuristically implies that our increment term dB_s has Hölder regularity $-\frac{1}{2}$. Hence if we fractionally integrate dB_s by order $H + \frac{1}{2}$ then we may result in a Gaussian process that is H -Hölder continuous which is exactly what we are looking for. Thus we may be tempted to formally define some process⁹

$$X_t = (I_0^{H+\frac{1}{2}} dB)(t) = \frac{1}{\Gamma(H+\frac{1}{2})} \int_0^t (t-s)^{H-\frac{1}{2}} dB_s.$$

However this does not have the stationary increments so cannot possibly be fBm . The process defined above was first introduced by Paul Lévy [11] and is known as Louiville Fbm or type 2 fBm. This process is explored more in [9] but this integral puts too great an importance on the origin for many applications.

In order to fix these issues it was suggested by Mandelbrot and Van-Ness in [11] that we should instead integrate from $-\infty$ to ensure that we have the stationary increments that we want, then subtract our Lévy representation to ensure that the process starts at the origin. Formally this suggests that fBm may be described by a stochastic process of the form:

$$\begin{aligned} X_t &= (I_{-\infty}^{H+\frac{1}{2}} dB)(t) - (I_0^{H+\frac{1}{2}} dB)(t) \\ &= \frac{1}{\Gamma(H+\frac{1}{2})} \left(\int_{-\infty}^0 \left((t-u)^{H-\frac{1}{2}} - (-u)^{H-\frac{1}{2}} \right) dB_u + \int_0^t (t-u)^{H-\frac{1}{2}} dB_u \right) \end{aligned}$$

In fact the above process is actually (up to scaling) fBm_H and turns out to be one of the most important representations.

Theorem 1.3.1 (Mandelbrot-Vann Ness / Time Representation of fBm_H [13]).

$$\bar{B}_t^H = \frac{1}{c_H} \left(\int_{-\infty}^0 \left((t-u)^{H-\frac{1}{2}} - (-u)^{H-\frac{1}{2}} \right) dB_u + \int_0^t (t-u)^{H-\frac{1}{2}} dB_u \right)$$

where

$$c_H = \sqrt{\frac{1}{2H} + \int_0^\infty \left((1+u)^{H-\frac{1}{2}} - u^{H-\frac{1}{2}} \right)^2 du}$$

is a fBm_H .

Proof. It is clear that an integral with respect to a Gaussian process is again a Gaussian process, and therefore $\bar{B}^H$ is also a Gaussian process. Furthermore we can see that $\mathbb{E}(\bar{B}_t^H) = 0$. Therefore, $\bar{B}^H$ is a centred Gaussian process.

⁹Note that the power term in the integrand is $H + \frac{1}{2} - 1 = H - \frac{1}{2}$.

Now, to show that $\bar{B}^H$ has stationary increments, we rely on the fact that the Wiener process has stationary and independent increments. So taking some $s > t$ we have

$$\begin{aligned}
(\bar{B}_t^H - \bar{B}_s^H) &= \frac{1}{c_H^2} \int_{\mathbb{R}} (t-u)_+^{H-\frac{1}{2}} - (s-u)_+^{H-\frac{1}{2}} dB_u \\
&= \frac{1}{c_H^2} \int_{\mathbb{R}} (t-s-v)_+^{H-\frac{1}{2}} - (-v)_+^{H-\frac{1}{2}} dB_{v+s} \quad (\text{where } v := u-s) \\
&\stackrel{d}{=} \frac{1}{c_H^2} \int_{\mathbb{R}} (t-s-v)_+^{H-\frac{1}{2}} - (-v)_+^{H-\frac{1}{2}} dB_v \quad (dB \text{ is stationary}) \\
&= (\bar{B}_{t-s}^H)
\end{aligned}$$

Therefore, $\bar{B}^H$ has stationary increments. Now it remains to show that $\bar{B}^H$ satisfies the covariance structure from 1.2.1.

Take $t > 0$, then by substituting $v := \frac{u}{t}$ for $t > 0$ and $v := 1 - \frac{u}{t}$ for $t < 0$ and factoring out our t terms along with using Ito isometry we get

$$\begin{aligned}
\mathbb{E}[(\bar{B}_t^H)^2] &= \frac{1}{c_H^2} \int_{\mathbb{R}} \left((t-u)_+^{H-\frac{1}{2}} - (-u)_+^{H-\frac{1}{2}} \right)^2 du \\
&= \frac{1}{c_H} |t|^{2H} \int_{\mathbb{R}} \left((1-v)_+^{H-\frac{1}{2}} - (-v)_+^{H-\frac{1}{2}} \right)^2 dv \quad (du = t dv) \\
&= \frac{1}{c_H} |t|^{2H} \left(\int_{-\infty}^0 \left((1-v)^{H-\frac{1}{2}} - (-v)^{H-\frac{1}{2}} \right)^2 dv + \int_0^1 (1-v)^{H-\frac{1}{2}} dv \right) \\
&= \frac{1}{c_H} |t|^{2H} \left(\int_{-\infty}^0 \left((1-v)^{H-\frac{1}{2}} - (-v)^{H-\frac{1}{2}} \right)^2 dv + \frac{1}{2H} \right) \\
&= |t|^{2H}
\end{aligned}$$

We then easily recover the required covariance since

$$\begin{aligned}
&\mathbb{E}[(B_t^H - B_s^H)^2] = |t-s|^{2H} \quad (\text{Since } \bar{B} \text{ is stationary}) \\
\implies \mathbb{E}[(B_t^H)^2] + \mathbb{E}[(B_s^H)^2] - 2\mathbb{E}[B_t^H B_s^H] &= |t-s|^{2H}. \\
\implies \text{Cov}(B_s^H, B_t^H) &= \mathbb{E}[B_s^H B_t^H] = \frac{1}{2} [s^{2H} + t^{2H} - |s-t|^{2H}].
\end{aligned}$$

□

This integral representation is itself the proof of the existence of fBm_H which was promised earlier. While the Mandelbrot-Van Ness representation is valuable both as a pedagogical tool demonstrating long-range dependence (see section 1.2.2) and for proving existence, it has a fundamental limitation. Since the integral runs over $(-\infty, t)$, the process B_t^H depends on the driving Brownian motion B_u for $u \in (-\infty, 0)$, meaning B_t^H is not adapted to the natural filtration $\mathcal{F}_t = \sigma(B_s, 0 \leq s \leq t)$. However standard techniques of martingale theory and Itô calculus requires adaptedness (see [17]) of precisely such a filtration, and so

cannot be directly applied to the Mandelbrot-Van Ness representation. To fix this, we instead seek a representation of the form:

$$B_t^H = \int_0^t K_H(t, s) dB_s$$

where K_H is a deterministic Volterra type kernel so that B_t^H depends only on W_s for $s \in [0, t]$ and adaptedness is recovered.

Theorem 1.3.2 (Volterra Kernel representation [13]). *Let $H \in (0, \frac{1}{2}) \cup (\frac{1}{2}, 1)$ and, for $t > s > 0$, set*

$$K_H(t, s) = \begin{cases} C_H^+ \cdot s^{\frac{1}{2}-H} \int_s^t (u-s)^{H-\frac{3}{2}} u^{H-\frac{1}{2}} du & \text{if } H > \frac{1}{2}, \\ C_H^- \cdot \left\{ \left(\frac{t}{s}\right)^{H-\frac{1}{2}} (t-s)^{H-\frac{1}{2}} - \left(H - \frac{1}{2}\right) s^{\frac{1}{2}-H} \int_s^t u^{H-\frac{3}{2}} (u-s)^{H-\frac{1}{2}} du \right\} & \text{if } H < \frac{1}{2}, \end{cases}$$

where

$$C_H^+ := \sqrt{\frac{H(2H-1)}{\int_0^1 (1-x)^{1-2H} x^{H-\frac{3}{2}} dx}}, \quad C_H^- := \sqrt{\frac{2H}{(1-2H) \int_0^1 (1-x)^{-2H} x^{H-\frac{1}{2}} dx}}.$$

Then the process

$$B_t^H = \int_0^t K_H(t, s) dW_s$$

is a fBm_H.

We choose to omit the lengthy proof of this theorem. This Volterra representation is crucial in the development of a theory of integration with respect to fBm - which will not be developed any further here (see [13]).

1.4 The Ornstein-Uhlenbeck Representation

As discussed, the non-Markovian nature of fBm makes it particularly difficult to work with, in particular Markovian process allow us to apply techniques such as a Feynman-Kac as well as efficient simulation. In this section we will show how we can lift fBm_H to a Markov process - a technique known as Markovian lifting [7]. In particular we will be lifting to an infinite dimensional Ornstein-Uhlenbeck process, thus it is important that we first establish such a process and demonstrate that it is indeed Markovian.

1.4.1 Ornstein-Uhlenbeck Process

The Ornstein-Uhlenbeck (OU) process $(U_t)_{t \geq 0}$ is a mean-reverting process defined by the following stochastic differential equation (SDE):

$$dU_t := -\beta(U_t - \mu)dt + \sigma dB_t.$$

Here μ is our expected value of the process; σ^2 is the variance; and β is our mean reversion rate. For the remainder of this chapter we will be working with the standard OU process where $\mu = 0$ and $\sigma = 1$ which we will denote by $(Z_t^\beta)_{t \geq 0}$ where

$$dZ_t^\beta := -\beta Z_t^\beta dt + dB_t.$$

Now by applying Ito's lemma [5] with the function $f(t, x) = xe^{\beta t}$ it follows that

$$\begin{aligned} d(Z_t^\beta e^{\beta t}) &= e^{\beta t} dB_t \\ \implies Z_t^\beta &= \int_{-\infty}^t e^{-\beta(t-u)} dB_u \end{aligned}$$

Where we choose to integrate from $-\infty$ to ensure that the process is stationary and that no single point is over emphasised. Notice also that Z_t^β is Gaussian with mean zero since it is a stochastic integral with deterministic kernel. Furthermore we can calculate the variance as follows:

$$\begin{aligned} \text{Var}(Z_t^\beta) &= \mathbb{E}[(Z_t^\beta)^2] = \mathbb{E} \left[\left(\int_{-\infty}^t e^{-\beta(t-u)} dB_u \right)^2 \right] \\ &= \int_{-\infty}^t e^{-2\beta(t-u)} du \quad (\text{By Ito isometry}) \\ &= \frac{1}{2\beta}. \end{aligned}$$

Thus we can simply write $Z_t^\beta \sim N(0, \frac{1}{2\beta})$. This makes sense as we think of Z_t^β as being a stationary process that is mean reverting, where the faster we are reverting to the mean (E.G larger β) would correspond to a smaller variance of the process.

1.4.2 Multivariate Ornstein-Uhlenbeck process

We can now generalize this process to be multi-dimensional, namely for arbitrary $m \in \mathbb{N}$, let $\boldsymbol{\beta} = (\beta_1, \dots, \beta_m)^T \in \mathbb{R}_+^m$, and $\mathbf{Z}_t^\beta = (Z_t^{\beta_1}, \dots, Z_t^{\beta_m})^T$. Following [12], we can describe $(\mathbf{Z}_t^\beta)_{t \geq 0}$ as a multivariate Ornstein-Uhlenbeck process with the following integral representation¹⁰:

$$\mathbf{Z}_t^\beta = \int_{-\infty}^t e^{-\text{diag}(\boldsymbol{\beta})(t-u)} I_m dB_u$$

The vector integrals are evaluated element-wise. Now, the transition densities for $(\mathbf{Z}_t^\beta)_{t \geq 0}$ will be required to show the Markov property, so we compute

¹⁰Here we use the standard definition of matrix exponentiation $e^A := \sum_{n=0}^{\infty} \frac{A^n}{n!}$

them here. Fix arbitrary $s, t \in \mathbb{R}_{\geq 0}$ such that $s < t$. Then

$$\begin{aligned} \mathbf{Z}_t^\beta &= \int_{-\infty}^t e^{-\text{diag}(\beta)(t-u)} \mathbf{1}_m dB_u \\ &= e^{-\text{diag}(\beta)(t-s)} \int_{-\infty}^s e^{-\text{diag}(\beta)(s-u)} I_m dB_u + \int_s^t e^{-\text{diag}(\beta)(t-u)} I_m dB_u \\ \mathbf{Z}_t^\beta &= e^{-\text{diag}(\beta)(t-s)} \mathbf{Z}_s^\beta + \int_s^t e^{-\text{diag}(\beta)(t-u)} I_m dB_u \end{aligned}$$

Notice that the second term is a multivariate Gaussian¹¹ hence the conditional distribution of $\mathbf{Z}_t^\beta \mid \mathbf{Z}_s^\beta$ is fully specified by its mean vector and covariance matrix (Since $\mathbf{Z}_s^\beta$ is a pre-determined constant vector in this case). Thus the mean is given by

$$\begin{aligned} \boldsymbol{\mu}_{t|s, \mathbf{Z}_s^\beta} &:= \mathbf{E} \left[\mathbf{Z}_t^\beta \mid \mathbf{Z}_s^\beta \right] = \mathbf{E} \left[e^{-\text{diag}(\beta)(t-s)} \mathbf{Z}_s^\beta \mid \mathbf{Z}_s^\beta \right] + \mathbf{E} \left[\int_s^t e^{-\text{diag}(\beta)(t-u)} I_m dB_u \mid \mathbf{Z}_s^\beta \right] \\ &= e^{-\text{diag}(\beta)(t-s)} \mathbf{Z}_s^\beta + \mathbf{0}_m = e^{-\text{diag}(\beta)(t-s)} \mathbf{Z}_s^\beta \end{aligned}$$

and the covariance matrix by

$$\begin{aligned} \Sigma_{t|s, \mathbf{Z}_s^\beta}^\beta &= \mathbf{E} \left[\left(\mathbf{Z}_t^\beta - \mathbf{E} \left[\mathbf{Z}_t^\beta \mid \mathbf{Z}_s^\beta \right] \right) \left(\mathbf{Z}_t^\beta - \mathbf{E} \left[\mathbf{Z}_t^\beta \mid \mathbf{Z}_s^\beta \right] \right)^T \mid \mathbf{Z}_s^\beta \right] \\ &= \mathbf{E} \left[\left(\int_s^t e^{-\text{diag}(\beta)(t-u)} \mathbf{1}_m dB_u \right) \left(\int_s^t e^{-\text{diag}(\beta)(t-u)} \mathbf{1}_m dB_u \right)^T \mid \mathbf{Z}_s^\beta \right] \\ &= \left[\mathbf{E} \left[\left(\int_s^t e^{-\beta_i(t-u)} dB_u \right) \left(\int_s^t e^{-\beta_j(t-u)} dB_u \right) \right] \right]_{i,j=1}^m \\ &= \left[\int_s^t e^{-\beta_i(t-u)} e^{-\beta_j(t-u)} du \right]_{i,j=1}^m \quad (\text{By Ito isometry}) \\ \Sigma_{t|s, \mathbf{Z}_s^\beta}^\beta &= \left[\frac{1 - e^{-(\beta_i + \beta_j)(t-s)}}{\beta_i + \beta_j} \right]_{i,j=1}^m \end{aligned}$$

Thus, $\mathbf{Z}_t^\beta \mid \mathbf{Z}_s^\beta \sim \mathcal{N} \left(\boldsymbol{\mu}_{t|s, \mathbf{Z}_s^\beta}, \Sigma_{t|s, \mathbf{Z}_s^\beta}^\beta \right)$ and hence has transition density:

$$f_{t|s, \mathbf{Z}_s^\beta}(\mathbf{z}) = \frac{1}{(2\pi)^{m/2} \sqrt{\det \left(\Sigma_{t|s, \mathbf{Z}_s^\beta}^\beta \right)}} \exp \left\{ -\frac{1}{2} \left(\mathbf{z} - \boldsymbol{\mu}_{t|s, \mathbf{Z}_s^\beta} \right)^T \Sigma_{t|s, \mathbf{Z}_s^\beta}^{-1} \left(\mathbf{z} - \boldsymbol{\mu}_{t|s, \mathbf{Z}_s^\beta} \right) \right\}$$

We next extend this notion to be infinite dimensional and demonstrate that it is indeed Markovian.

¹¹The second term is a vector of Ito integrals with respect to a Brownian motion, hence is a multivariate Normal

1.4.3 Infinite Dimensional OU process & the Markov Property

In the next section we will want to consider an OU process where the mean reversion speed can be any value in the continuum $\beta \in (0, \infty)$. Thus we must make sense of what such an infinite dimensional extension would look like¹². Thus suppose $t > 0$ is fixed and consider the “infinite dimensional vector” $\mathbf{Z}_t := (Z_t^\beta)_{\beta > 0}$ which we will denote by Z_t to emphasize the fact that β varies. It is demonstrated in [16] that this infinite dimensional process over all possible values of β actually exists and that we can define a sensible notion of probability distributions over such a space.

Now, to show that Z_t is Markovian, it is shown in [16] that it suffices to show the Markov property holds jointly on every finite-dimensional “cylinder”, i.e. for every $m \in \mathbb{N}$ and every finite grid $\beta = (\beta_1, \dots, \beta_m)^T \in \mathbb{R}_+^m$, the vector $\mathbf{Z}_t^\beta = (Z_t^{\beta_1}, \dots, Z_t^{\beta_m})^T$ must satisfy

$$\mathbf{Z}_t^\beta \mid \mathcal{F}_s \stackrel{d}{=} \mathbf{Z}_t^\beta \mid \mathbf{Z}_s^\beta.$$

Crucially, this must hold jointly across the coordinates $\beta_1, \dots, \beta_m$ for an arbitrary finite grid - showing the property coordinate-by-coordinate for each β_i in isolation is not sufficient, since it would say nothing about the joint dependence structure between coordinates. We now fix an arbitrary $m \in \mathbb{N}$ and grid $\beta \in \mathbb{R}_+^m$ and prove exactly this joint statement, drawing on the multivariate OU process established above.

Since random variables are uniquely determined by their characteristic functions [5], it suffices to show that the corresponding characteristic functions agree, as follows.

$$\begin{aligned} \mathbf{E} \left[e^{i\mathbf{x}' \mathbf{Z}_t^\beta} \mid \mathcal{F}_s \right] &= \mathbf{E} \left[\exp \left\{ i\mathbf{x}' \left(e^{-\text{diag}(\beta)(t-s)} \mathbf{Z}_s^\beta + \int_s^t e^{-\text{diag}(\beta)(t-u)} \mathbf{1}_m dB_u \right) \right\} \mid \mathcal{F}_s \right] \\ &= \exp \left\{ i\mathbf{x}' \left(e^{-\text{diag}(\beta)(t-s)} \mathbf{Z}_s^\beta \right) \right\} \mathbf{E} \left[\exp \left\{ i\mathbf{x}' \left(\int_s^t e^{-\text{diag}(\beta)(t-u)} \mathbf{1}_m dB_u \right) \right\} \mid \mathcal{F}_s \right] \\ &= \exp \left\{ i\mathbf{x}' \left(e^{-\text{diag}(\beta)(t-s)} \mathbf{Z}_s^\beta \right) \right\} \mathbf{E} \left[\exp \left\{ i\mathbf{x}' \left(\int_s^t e^{-\text{diag}(\beta)(t-u)} \mathbf{1}_m dB_u \right) \right\} \mid \mathbf{Z}_s^\beta \right] \\ &= \mathbf{E} \left[\exp \left\{ i\mathbf{x}' \left(e^{-\text{diag}(\beta)(t-s)} \mathbf{Z}_s^\beta + \int_s^t e^{-\text{diag}(\beta)(t-u)} \mathbf{1}_m dB_u \right) \right\} \mid \mathbf{Z}_s^\beta \right] \\ &= \mathbf{E} \left[e^{i\mathbf{x}' \mathbf{Z}_t^\beta} \mid \mathbf{Z}_s^\beta \right]. \end{aligned}$$

For the third equality, we used the fact that the random variable in the expectation is independent of both $\mathcal{F}_s$ and $\mathbf{Z}_s^\beta$. Characteristic functions completely specify their corresponding distributions, so $\mathbf{Z}_t^\beta \mid \mathcal{F}_s \stackrel{d}{=} \mathbf{Z}_t^\beta \mid \mathbf{Z}_s^\beta$ as required.

¹²In this section we are deliberately somewhat non-rigorous in defining such an infinite dimensional process as we want to stick to the main idea of this project and not get bogged down in the measure theoretic details (see [16]).

Since $m \in \mathbb{N}$ and the grid $\beta \in \mathbb{R}_+^m$ were arbitrary, the above shows that the Markov property holds jointly on every finite-dimensional cylinder. Hence, by [16], the infinite dimensional Z_t is itself Markovian, as required.

1.4.4 Derivation of Muravlev Representation

Now that we have built up the required understanding of an Ornstein-Uhlenbeck process and extended it to infinitely many dimensions, we are ready to fully appreciate how it can be used to represent fBm $_H$. The derivation essentially comes down to algebraic trickery using the Gamma function, however a full appreciation of the importance requires our developed knowledge of the OU process from the last section. The next theorem is the key algebraic trick that allows us to develop the so called Muravlev representation.

Theorem 1.4.1 (Ornstein-Uhlenbeck representation of the fractional kernel). *For $\tau \in \mathbb{R}_{>0}$ and $H \in (0, 1/2) \cup (1/2, 1)$ we have the following formulae.*

$$\tau^{H-1/2} \Gamma(1/2 - H) = \begin{cases} \int_0^\infty \beta^{-1/2-H} e^{-\tau\beta} d\beta, & \text{if } H \in (0, 1/2) \\ \int_0^\infty \beta^{-1/2-H} (e^{-\tau\beta} - 1) d\beta, & \text{if } H \in (1/2, 1) \end{cases}$$

Proof. We present here a sketch of that given by [16]. We begin with the easier case $H \in (0, 1/2)$, which simply follows from the known Laplace transform identity in which $\tau \mapsto \tau^{H-1/2}$ is the Laplace transform of the function $\beta \mapsto \beta^{-1/2-H} / \Gamma(1/2 - H)$. This can also easily be shown from scratch using the substitution $u = \beta\tau$ which we do not present here.

Now the $H \in (1/2, 1)$ case is a bit more intricate. In fact the integral definition of $\Gamma(x)$ does not converge for $x < 0$, however (noting that we want this to extend the factorial function) the definition can be extended via the identity $x\Gamma(x) = \Gamma(x+1)$. In particular, for $x \in (-1, 0)$ one has

$$\begin{aligned} \Gamma(x) &= \frac{1}{x} \Gamma(x+1) = \frac{1}{x} \int_0^\infty u^x e^{-u} du \\ &= \frac{1}{x} \left([-u^x (e^{-u} - 1)]_{u=0}^\infty + \int_0^\infty x u^{x-1} (e^{-u} - 1) du \right) = \int_0^\infty u^{x-1} (e^{-u} - 1) du \end{aligned}$$

using integration by parts. Then substituting $u = \beta\tau$ into the integral on the right-hand side of gives we see that

$$\begin{aligned} \int_0^\infty \beta^{-1/2-H} (e^{-\beta\tau} - 1) d\beta &= \tau^{H-1/2} \int_0^\infty u^{(1/2-H)-1} (e^{-u} - 1) du \\ &= \tau^{H-1/2} \Gamma(1/2 - H) \end{aligned}$$

□

We now derive the Muravlev preservation by simply applying this trick to the Mandelbrot-Vann Ness representation.

Theorem 1.4.2 (Muravlev representation). *Let $H \in (0, 1/2) \cup (1/2, 1)$. For each $\beta \in \mathbb{R}_{>0}$, let $(Z_t^\beta)_{t \geq 0}$ be the standard Ornstein-Uhlenbeck process. If*

$$\bar{B}_t^H = \begin{cases} \frac{1}{\bar{c}_H} \int_0^\infty \beta^{-1/2-H} \left(Z_t^\beta - Z_0^\beta \right) d\beta, & \text{if } H \in (0, 1/2) \\ \frac{1}{\bar{c}_H} \int_0^\infty \beta^{-1/2-H} \left(Z_t^\beta - Z_0^\beta - B_t \right) d\beta, & \text{if } H \in (1/2, 1), \end{cases}$$

where the constant

$$\bar{c}_H = c_H \times \Gamma\left(\frac{1}{2} - H\right) = \Gamma\left(\frac{1}{2} - H\right) \sqrt{\frac{1}{2H} + \int_0^\infty \left((1+u)^{H-\frac{1}{2}} - u^{H-\frac{1}{2}} \right)^2 du}$$

then $(\bar{B}_t^H)_{t \in \mathbb{R}_{\geq 0}}$ is a fBm_H .

Proof. We start with the Mandelbrot-Van Ness representation. For $t \in \mathbb{R}_{\geq 0}$,

$$\begin{aligned} B_t^H &= \frac{1}{c_H} \int_{\mathbb{R}} \left((t-s)_+^{H-1/2} - (-s)_+^{H-1/2} \right) dB_s. \\ &= \frac{1}{c_H} \int_{-\infty}^0 \left((t-s)^{H-1/2} - (-s)^{H-1/2} \right) dB_s + \frac{1}{c_H} \int_0^t (t-s)^{H-1/2} dB_s. \end{aligned}$$

Applying Theorem 1.4.1 to each power $(t-s)^{H-1/2}$ and $(-s)^{H-1/2}$, and noting that $\bar{c}_H = c_H \Gamma(1/2 - H)$, that gives us the first case $H \in (0, \frac{1}{2})$

$$\begin{aligned} B_t^H &= \frac{1}{\bar{c}_H} \int_{-\infty}^0 \left[\int_0^\infty \beta^{-1/2-H} \left(e^{-\beta(t-s)} - e^{\beta s} \right) d\beta \right] dB_s \\ &\quad + \frac{1}{\bar{c}_H} \int_0^t \left[\int_0^\infty \beta^{-1/2-H} e^{-\beta(t-s)} d\beta \right] dB_s. \end{aligned}$$

and for the case when $H \in (1/2, 1)$:

$$\begin{aligned} B_t^H &= \frac{1}{\bar{c}_H} \int_{-\infty}^0 \left[\int_0^\infty \beta^{-1/2-H} \left(e^{-\beta(t-s)} - e^{\beta s} \right) d\beta \right] dB_s \\ &\quad + \frac{1}{\bar{c}_H} \int_0^t \left[\int_0^\infty \beta^{-1/2-H} \left(e^{-\beta(t-s)} - 1 \right) d\beta \right] dB_s. \end{aligned}$$

Now using a stochastic Fubini theorem (see [5]) to interchange the order of integration and recalling,

$$Z_t^\beta := \int_{-\infty}^t e^{-\beta(t-s)} dB_s$$

we get

$$B_t^H = \begin{cases} \frac{1}{\bar{c}_H} \int_0^\infty \beta^{-1/2-H} \left(Z_t^\beta - Z_0^\beta \right) d\beta, & \text{if } H \in (0, 1/2), \\ \frac{1}{\bar{c}_H} \int_0^\infty \beta^{-1/2-H} \left(Z_t^\beta - Z_0^\beta - B_t \right) d\beta, & \text{if } H \in (1/2, 1), \end{cases}$$

which is Muravlev's OU representation. $\square$

This representation is much easier to work with as it allows us to instead work with Z_t which is easier to both simulate and work with analytically. Moreover it provides Heuristic insights as to the inner workings of fBm_H as we can think of it as simply a linear combination of OU process' with different mean reversions. We next use this representation to approximate fBm_H .

1.5 Simulating fBm

Now that we have seen all of the basic theory, the natural next step is the implementation of fBm_H simulation. The fact that fBm_H is non-Markovian makes simulation methods difficult both in terms of the mathematics and computational cost. In this section we will present the Cholesky method for simulating fBm_H where we essentially just simulate the covariance between all increments. In section 1.6 we will demonstrate another simulation method that makes use of both the Cholesky method and also an integral representation.

1.5.1 Cholesky Method

The Cholesky method is a standard tool for simulating Gaussian processes. $A \in \mathbb{R}^{m \times n}$ is a matrix, $\mu \in \mathbb{R}^n$ a vector then we can define $\mathbf{X} = \mu + A\mathbf{Z}$. Then basic manipulation shows that $\mathbf{X}$ has a multivariate Gaussian distribution with covariance matrix $\Sigma = AA^T$ and mean $\mu = \mathbb{E}[\mathbf{X}]$.

Recall that a positive semi-definite matrix Σ has a Cholesky decomposition $\Sigma = LL^T$, where L is a lower triangular matrix. Thus by the previous paragraph and the fact that all covariance matrices are positive semi-definite, we therefore can formulate the following algorithm for simulating a multivariate Gaussian.

Algorithm 1

Input:

- $\Sigma \in \mathbb{R}^{n \times n}$ positive definite covariance matrix
- $\mu \in \mathbb{R}^n$ mean vector
- $m \in \mathbb{N}$ number of samples

Output:

- m samples of multivariate Gaussian with mean μ and covariance Σ .

1. Compute L such that $\Sigma = LL^T$.
 2. Repeat the following m times:
 - a) Simulate n independent samples from the standard normal distribution $\mathbf{Z} = (Z_1, \dots, Z_n)$.
 - b) Save new sample $\mu + L\mathbf{Z}$.
-

To implement this algorithm we further need to understand how to proceed with steps 1. and 2.a). The simulation of independent normal random variables is beyond the scope of this report¹³. Regarding the computation of the Cholesky

¹³In Python we can use the numpy package and the command `numpy.random.randn(n)`.

decomposition, one can use the fact that L is lowertriangular to compute its entries iteratively from the top to the bottom. Here is a simple Python code that works for positive definite matrices:

```

1 import numpy as np
2
3 def cholesky(A):
4     L = np.zeros_like(A)
5     n, _ = A.shape
6     for i in range(n):
7         for j in range(i + 1):
8             sum_ = np.sum(L[i] * L[j])
9             if i == j:
10                L[i, j] = np.sqrt(A[i, i] - sum_)
11            else:
12                L[i, j] = (A[i, j] - sum_) / L[j, j]
13     return L

```

Note that the costs of this algorithm of the order $O(n^3)$ (a third inner loop is hidden in the numpy sum).

We now consider fBm_H which is a centred Gaussian process with $C(t, s) = Cov(X_t, X_s) = \frac{1}{2}[t^{2H} + s^{2H} - |t - s|^{2H}]$. Hence if we use n time steps over a time intervals T we can discretize on a time grid $\{0 = t_0 < t_1 < \dots < t_n = T\}$ which then simply amounts to the simulation of the $(n + 1)$ -dimensional Gaussian random variable $\mathbf{X} := (X_{t_0}, \dots, X_{t_n})$ which we can write as $\mathbf{X} = \Sigma \mathbf{Z}$ where $(\Sigma)_{ij} = C(t_i, t_j)$, we then simply apply the Cholesky algorithm ¹⁴

1.6 Markovian Lifting

We will next apply this OU representation to consider an approximation of fBm_H . We follow the ideas of [7], in which a stochastic Volterra process with the Liouville fractional kernel is approximated (we dedicate Chapter 2 to this model); they achieve this by approximating the fractional kernel using only finitely many mean-reversion terms. Our aim here is to adapt their method to approximate true fBm_H directly - hereafter we refer to this approximation as *lifted* fBm_H .

1.6.1 Fractional Kernel Approximation for $H < \frac{1}{2}$

We begin with a simple approximation of the fractional kernel, which is central to the OU representation. Recall

$$K_H(t, u) = \frac{(t - u)^{H - \frac{1}{2}}}{\Gamma(H + \frac{1}{2})} = \frac{1}{\Gamma(\frac{1}{2} - H) \Gamma(H + \frac{1}{2})} \int_0^\infty \beta^{-\frac{1}{2} - H} e^{-(t - u)\beta} d\beta.$$

Which we may rewrite as

$$K_H(t, u) = \int_0^\infty e^{-(t - u)\beta} \mu(d\beta) \quad \text{where} \quad \mu(d\beta) := \frac{\beta^{-\frac{1}{2} - H}}{\Gamma(\frac{1}{2} - H) \Gamma(H + \frac{1}{2})} d\beta.$$

¹⁴A full Python implementation can be found here to simulate the process.

We approximate the kernel by discretizing this integral:

$$K^n(t, u) = \sum_{j=-n+1}^n w_j e^{-(t-u)\beta_j}$$

for weights w_j to be determined. Following [7], we use a geometric quadrature for fixed $n \in \mathbb{N}$ with $\beta_j = r^j$ for some $r \in (1, 2)$ and $j = -n+1, \dots, n$. Heuristically, larger values of β correspond to faster mean reversion and hence a shorter time scale, so they contribute less to the overall process - this motivates spacing the β_j geometrically, concentrating more nodes near the slower (small- β) modes that dominate the long-memory behaviour.

To obtain the weights¹⁵, we hold $e^{-(t-u)\beta}$ constant for $\beta_j = r^j$ on each subinterval $[r^{j-1}, r^j]$ and integrate:

$$w_j = \frac{1}{\Gamma(\frac{1}{2} - H) \Gamma(H + \frac{1}{2})} \int_{r^{j-1}}^{r^j} \beta^{-\frac{1}{2}-H} d\beta.$$

Evaluating the integral gives

$$w_j = \frac{1}{\Gamma(\frac{1}{2} - H) \Gamma(H + \frac{1}{2})} r^{(j-1)(\frac{1}{2}-H)} \left[\frac{r^{\frac{1}{2}-H} - 1}{\frac{1}{2} - H} \right].$$

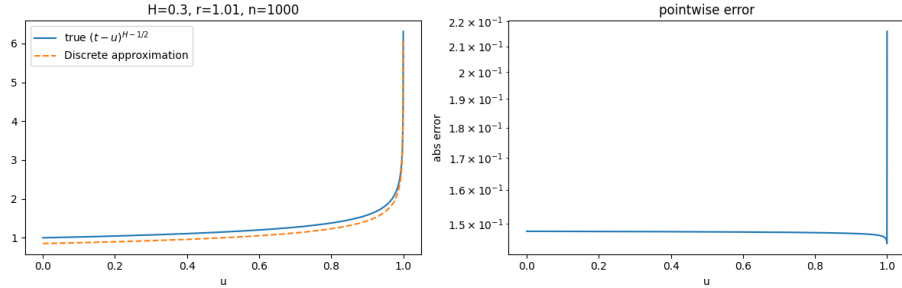


Figure 1.2: Comparison of the true fractional kernel for $t = 1$ such that $K(t, u)$ (blue, solid) with the lifted approximation $\sum_{j=-n+1}^n w_j e^{-(t-u)\beta_j}$ (orange, dashed), for $H = 0.3$, $r = 1.01$, $n = 1000$ and $t = 1$. The right panel shows the relative approximation error. The approximation is accurate over an intermediate range of lags but degrades near $u = t$ reflecting the singularity of the fractional kernel causing large blow-ups.

¹⁵A convergence proof of $K^n \rightarrow K_H$ in the L^2 sense is expected to follow along similar lines to Appendix A3 of [7], though the underlying quadrature there differs from the simple geometric scheme used here (cf. the weights in (2.9)). We do not pursue a formal proof and instead provide empirical validation via Monte Carlo in the sections that follow.

1.6.2 Lifted fBm for $H < \frac{1}{2}$

We now extend this same discretization method to approximate fBm_H . Recall from Section 1.4.4 that for $H < \frac{1}{2}$ we can represent fBm_H by

$$B_t^H = \frac{1}{\bar{c}_H} \int_0^\infty \beta^{-1/2-H} \left(Z_t^\beta - Z_0^\beta \right) d\beta, \quad (1.1)$$

where $\bar{c}_H$ is a constant and, for each $\beta \in (0, \infty)$, Z_t^β is an OU process with mean reversion β .

We can now use this representation to obtain an approximate Markovian simulation method. As before, we discretize the integral:

$$B_t^{H,n} = \sum_{j=-n+1}^n w_j \left(Z_t^{\beta_j} - Z_0^{\beta_j} \right).$$

In the same way as in Section 1.6.1, to obtain the weights we hold $Z_t^\beta - Z_0^\beta$ fixed at $\beta = \beta_j = r^j$ on each subinterval $[r^{j-1}, r^j]$ and integrate the remaining term:

$$w_j = \frac{1}{\bar{c}_H} \int_{r^{j-1}}^{r^j} \beta^{-\frac{1}{2}-H} d\beta.$$

Evaluating the integral gives

$$w_j = \frac{1}{\bar{c}_H} r^{(j-1)(\frac{1}{2}-H)} \left[\frac{r^{\frac{1}{2}-H} - 1}{\frac{1}{2} - H} \right].$$

Thus we have the required weights to implement our lifted fBm_H . We note that standard methods can be used to simulate the OU process Z_t^β ; see Appendix A.

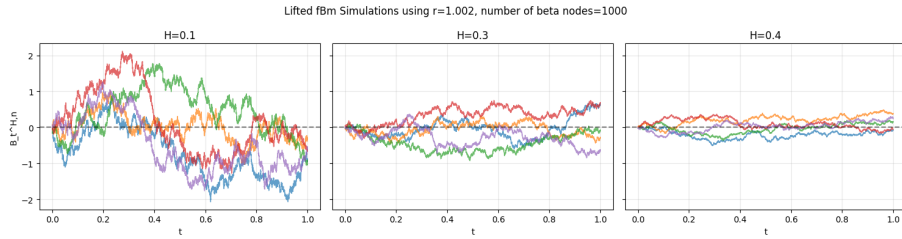


Figure 1.3: A Python implementation of lifted fBm_H for $H = 0.1, 0.3, 0.4$, using $r = 1.002$, $n = 500$. This aligns with what we expect: smaller values of H produce paths with larger local variance that revert toward the origin more quickly, giving a rougher-looking trajectory, while larger H produces smoother, less varied paths.

In order to validate that these paths do in fact approximate true fBm_H , we use Monte Carlo methods to estimate the covariance structure and compare it

against the known covariance of fBm. Specifically, we simulate $M = 5000$ sample paths over an evenly spaced grid of $N = 200$ points $\{0 = t_0 < t_1 < \dots < t_N = 1\}$ and estimate the covariance matrix as follows. Our desired covariance structure is given by

$$C(i, j) = \mathbb{E} \left[B_{t_i}^H B_{t_j}^H \right] = \frac{1}{2} \left[t_i^{2H} + t_j^{2H} - |t_i - t_j|^{2H} \right].$$

Now, defining $\mathbf{B} \in \mathbb{R}^{M \times N}$ as the matrix whose m -th row is the m -th simulated path evaluated at the grid points $t_0, \dots, t_N$, the Monte Carlo estimate of the covariance matrix is

$$\hat{C}_{(i,j)} = \frac{1}{M} \sum_{m=1}^M \mathbf{B}(m, i) \cdot \mathbf{B}(m, j) = \frac{1}{M} (\mathbf{B}^\top \mathbf{B}) [i, j].$$

By the law of large numbers, $\hat{C}$ converges to the true covariance C as $M \rightarrow \infty$, so for M sufficiently large $\hat{C}$ gives a reliable empirical picture of the covariance structure of our lifted fBm.

To validate the lifted approximation, we examine two quantities. First, to assess how the approximation improves as more mean-reversion terms are added, we vary n (shrinking r correspondingly to $r = 1 + \frac{1}{n}$) and plot the maximum covariance error $\max_{i,j} |\hat{C}(i, j) - C(i, j)|$ against n . Second, for a fixed n , we plot a heatmap of the relative covariance error,

$$\left| \frac{C(i, j) - \hat{C}(i, j)}{C(i, j)} \right|.$$

We highlight two key features of the right panel of Figure 1.4: The First being the large error along the diagonal $s \approx t$. This follows from the singularity of the fractional kernel: We saw in 1.2 that the fractional kernel $K(t, s)$ is badly approximated for $t \approx s$, recall that in the derivation for this representation of fBm we used the fractional kernel which therefore is implicit in the formula we are using. So our discrete approximation is least accurate near the kernel singularities $u = s$ and $u = t$. When $s \approx t$, these two poorly-approximated regions coincide, compounding the error and producing the pronounced ridge along the diagonal.

We next notice the larger relative error for small s or t . This is a normalization artefact rather than a failure of the approximation: the true covariance itself is small when s or t is close to 0, so even a small absolute error is amplified into a large *relative* error.

1.6.3 Fractional Kernel Approximation for $H > \frac{1}{2}$

For $H > \frac{1}{2}$, the representation used in Section 1.6.1 breaks down: the integral $\int_0^\infty \beta^{-\frac{1}{2}-H} e^{-(t-u)\beta} d\beta$ no longer converges near $\beta = 0$ once $H > \frac{1}{2}$. Instead we

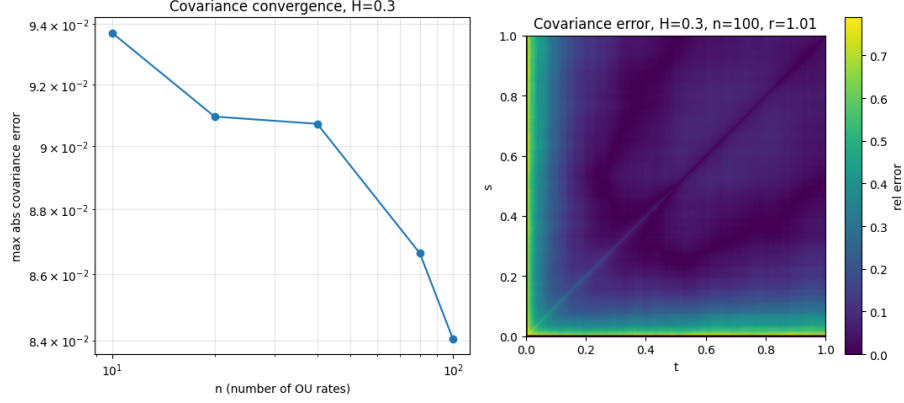


Figure 1.4: Left panel: the maximum covariance error, which decreases as more mean-reversion terms n are added. Right panel: relative error between the Monte Carlo covariance estimate $\hat{C}$ and the true fBm covariance, as a function of $(s, t) \in [0, T]^2$.

use the modified representation derived in 2.6:

$$K_H(t, u) = \frac{(t-u)^{H-\frac{1}{2}}}{\Gamma(H+\frac{1}{2})} = \frac{1}{\Gamma(\frac{1}{2}-H)\Gamma(H+\frac{1}{2})} \int_0^\infty \beta^{-\frac{1}{2}-H} \left(e^{-(t-u)\beta} - 1 \right) d\beta,$$

where subtracting 1 inside the integrand restores convergence¹⁶ at $\beta = 0$. As before, we may write this as

$$K_H(t, u) = \int_0^\infty \left(e^{-(t-u)\beta} - 1 \right) \mu(d\beta) \quad \text{where} \quad \mu(d\beta) := \frac{\beta^{-\frac{1}{2}-H}}{\Gamma(\frac{1}{2}-H)\Gamma(H+\frac{1}{2})} d\beta.$$

We approximate the kernel by discretizing this integral in the same way as Section 1.6.1:

$$K^n(t, u) = \sum_{j=-n+1}^n w_j \left(e^{-(t-u)\beta_j} - 1 \right)$$

using the same geometric quadrature $\beta_j = r^j$, $r \in (1, 2)$, $j = -n+1, \dots, n$, for the same reasons as before: larger β corresponds to faster mean reversion and a shorter time scale, motivating a geometric spacing that concentrates nodes near the slower, long-memory-dominating modes.

Holding $e^{-(t-u)\beta} - 1$ fixed at $\beta = \beta_j = r^j$ on each subinterval $[r^{j-1}, r^j]$ and integrating the remaining term gives

$$w_j = \frac{1}{\Gamma(\frac{1}{2}-H)\Gamma(H+\frac{1}{2})} \int_{r^{j-1}}^{r^j} \beta^{-\frac{1}{2}-H} d\beta,$$

¹⁶As $\beta \rightarrow 0$, $e^{-(t-u)\beta} - 1 \sim -(t-u)\beta$, so the integrand behaves like $\beta^{\frac{1}{2}-H}$, which is integrable near 0 for $H < \frac{3}{2}$.

which evaluates to

$$w_j = \frac{1}{\Gamma\left(\frac{1}{2} - H\right) \Gamma\left(H + \frac{1}{2}\right)} r^{(j-1)\left(\frac{1}{2} - H\right)} \left[\frac{r^{\frac{1}{2} - H} - 1}{\frac{1}{2} - H} \right].$$

Note this is the same weight formula as in the $H < \frac{1}{2}$ case (Section 1.6.1) - the correction term only enters through the -1 inside the sum, not through the weights themselves.

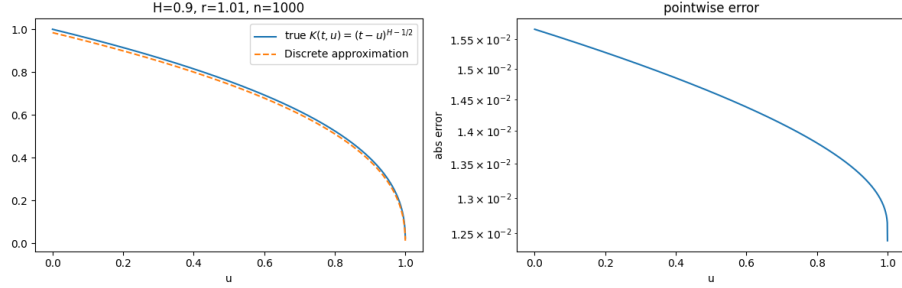


Figure 1.5: Comparison of the true fractional kernel for $t = 1$, $K(t, u) = (1 - u)^{H - \frac{1}{2}} / \Gamma(H + \frac{1}{2})$ (blue, solid), with the lifted approximation $K^n(t, u) = \sum_{j=-n+1}^n w_j (e^{-(t-u)\beta_j} - 1)$ (orange, dashed), for $H = 0.9$, $r = 1.01$, $n = 1000$, $t = 1$. The right panel shows the relative approximation error. This case exhibits a key qualitative difference from the $H < \frac{1}{2}$ regime: since $H - \frac{1}{2} > 0$, the kernel $(t - u)^{H - \frac{1}{2}}$ vanishes rather than blows up as $u \rightarrow t$, so there is no singularity to approximate near the diagonal, and correspondingly no error blow-up there.

1.6.4 Lifted fBm for $H > \frac{1}{2}$

We now extend the discretization method to $H > \frac{1}{2}$. Recall from Section 1.4.4 that in this regime fBm $_H$ admits the representation

$$B_t^H = \bar{c}_H \int_0^\infty \beta^{-1/2-H} \left(Z_t^\beta - Z_0^\beta - B_t \right) d\beta, \quad (1.2)$$

where B_t is a standard Brownian motion driving the same noise as each Z_t^β , and $\bar{c}_H$ is a constant. The extra term $-B_t$, absent in the $H < \frac{1}{2}$ representation (1.1), plays the same role as the -1 correction inside the integrand of the fractional kernel in Section 1.6.3: it is what restores convergence of the β -integral once $H > \frac{1}{2}$.

Discretizing as before,

$$B_t^{H,n} = \sum_{j=-n+1}^n w_j \left(Z_t^{\beta_j} - Z_0^{\beta_j} \right) - \left(\sum_{j=-n+1}^n w_j \right) B_t \quad (1.3)$$

using exactly the same weights w_j derived in Section 1.6.3. Since B_t does not depend on j , it factors out of the sum with total weight $\sum_j w_j$, so no new quadrature is needed for this term - we simulate B_t as the shared driving Brownian motion underlying each $Z_t^{\beta_j}$, exactly as in Appendix A.

Naively reusing the same geometric grid $\beta_j = r^j$ that worked well for $H < \frac{1}{2}$ does not carry over to this regime, however. The node range $[r^{-n+1}, r^n]$ is fixed once (r, n) are chosen, and for r close to 1 — needed for fine local resolution of the density reaching large β requires a correspondingly large n . Under a modest, practical (r, n) budget, the grid fails to reach β large enough to represent the fast-mean-reverting components of (1.2). These fast components are precisely what generate the short time-scale, rough-looking increments, so losing them produces artificially smooth sample paths, as seen below for $H = 0.51, 0.6, 0.9$.

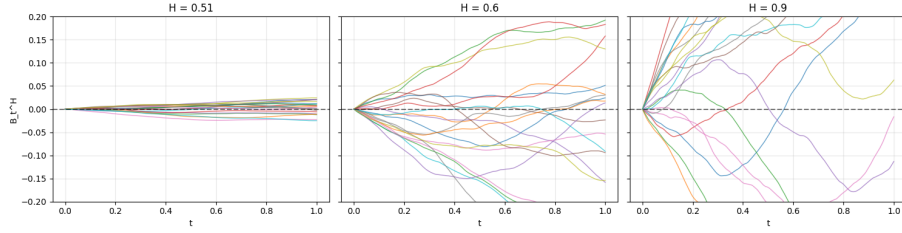


Figure 1.6: Sample paths of lifted fBm_H for $H = 0.51, 0.6, 0.9$ using the geometric grid from Section 1.6.3. The paths are visibly too smooth, suggesting the fast-mean-reverting terms are not being captured.

This failure mode is specific to $H > \frac{1}{2}$: for $H < \frac{1}{2}$, the long-memory behaviour is dominated by *small* β , and the geometric grid's natural concentration of nodes near small β works in its favour rather than against it. For $H > \frac{1}{2}$ we need the opposite - reliable coverage of large β - so instead of letting the node range fall out of (r, n) , we choose it directly from the time-scales the simulation can actually resolve:

$$\beta_{\min} = \frac{1}{T}, \quad \beta_{\max} = \frac{1}{dt},$$

where $dt = T/N_{\text{steps}}$ is the simulation time step, and distribute n nodes evenly in $\log \beta$ across $[\beta_{\min}, \beta_{\max}]$. The corresponding weights are again the exact mass of $\beta^{-\frac{1}{2}-H}/\Gamma(\frac{1}{2}-H)$ over each cell $[\beta_{i-1}, \beta_i]$, with the first and last cells absorbing the tail mass below $\beta_{\min}$ and above $\beta_{\max}$ respectively. Because $\beta_{\max} = 1/dt$ is reached by construction regardless of n , the fast-mean-reverting terms are always included, without the delicate (r, n) tuning the geometric grid would otherwise require.

As before, we validate against the known covariance

$$C(i, j) = \frac{1}{2} [t_i^{2H} + t_j^{2H} - |t_i - t_j|^{2H}]$$

using $M = 5000$ Monte Carlo paths.

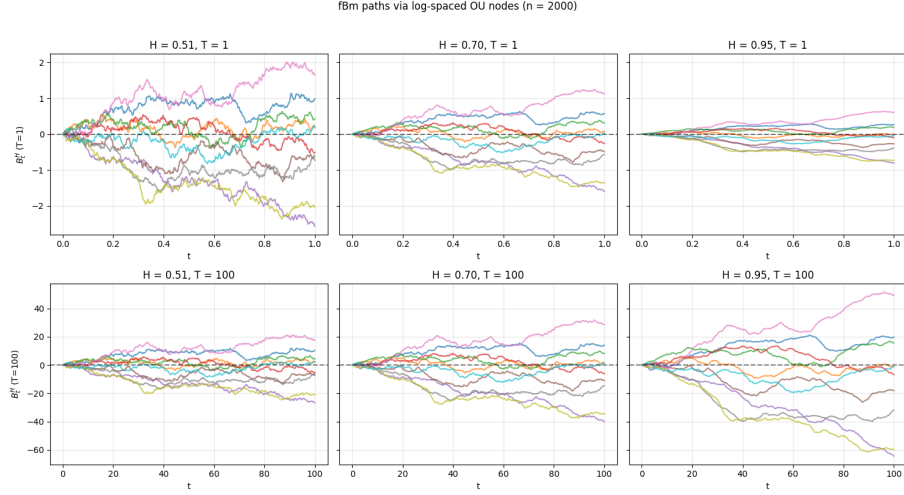


Figure 1.7: Sample paths of lifted fBm_H for $H = 0.51, 0.7, 0.95$, simulated using the log-spaced grid with $n = 2000$. On the top panel ($T = 1$) the larger H values are visibly smoother; on the bottom panel ($T = 100$) the larger H values show much more correlation between consecutive increments (“momentum”) and tend to drift away from the mean, as expected.

Together, Figure 1.8 gives empirical evidence that our lifted fBm (1.7) is broadly consistent with the covariance structure of fBm_H for $H > 1/2$, though a non-trivial residual error remains, particularly as H approaches 1 and near the boundary of $[0, T]^2$. We attribute this residual, at least in part, to the treatment of the low-frequency ($\beta \rightarrow 0$) end of the quadrature grid: for $H > \frac{1}{2}$ the spectral density $\beta^{-1/2-H}$ underlying the OU superposition is not integrable near $\beta = 0$, so the mass below the smallest grid node $\beta_{\min}$ must be handled with care, and we found this boundary treatment to be delicate rather than a fully settled part of the construction. A more careful choice of $\beta_{\min}$ and $\beta_{\max}$, and a rigorous L^2 convergence proof analogous to Theorem 2.5.3 for the lifted Heston model, are natural directions for future refinement; see also the discussion in Chapter 3.

The importance of lifted fBm_H extends well beyond a numerical curiosity. The key structural fact is that the process is fully determined by the finite collection of OU rates $(Z_t^{\beta_1}, \dots, Z_t^{\beta_n})$, and this finite-dimensional system is Markovian unlike fBm_H itself. This unlocks the full toolbox of techniques that apply only to Markov processes: infinitesimal generators, Feynman-Kac representations, and PDE-based characterizations, none of which are available for the non-Markovian, non-semimartingale process B_t^H directly. This is precisely the idea that motivates the *lifted Heston model*, to which we dedicate Chapter 2: replacing the non-Markovian fractional kernel driving the rough Heston variance process with the same finite superposition of OU processes developed

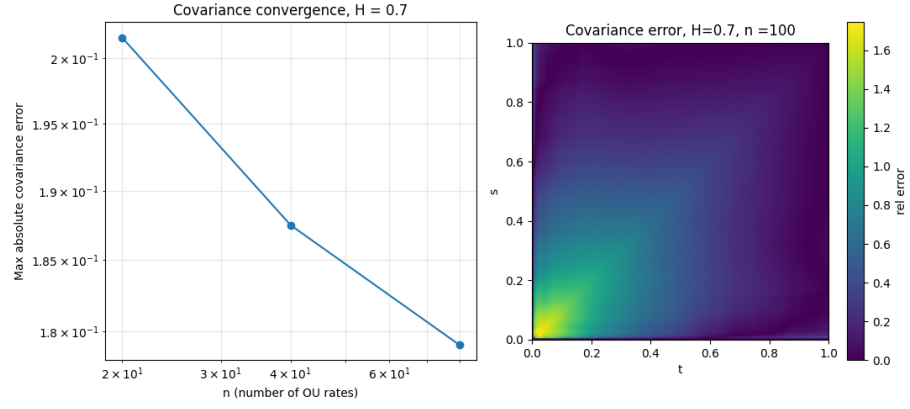


Figure 1.8: Relative covariance error for $H = 0.7$, $M = 5000$, $n = 100$, as a function of $(s, t) \in [0, T]^2$.

here, yielding a tractable, genuinely Markovian approximation.

Chapter 2

Rough Volatility Modelling

In this chapter we give a rapid introduction to volatility modelling, to give the reader a sense of how the same ideas from our lifted fBm_H is used in the lifted Heston model.

Now that we have developed the theory of fractional Brownian motion, we are ready to apply it to a modern and active area of volatility modelling. We begin this chapter with the classical Black-Scholes model, which we show is insufficient and unrealistic as a description of how option prices behave in practice. This motivates stochastic volatility models, and in particular the Heston model, whose relatively simple characteristic function alongside realistic fits for the implied volatility surface have made it one of the most popular models in the literature.

However, in [6] the authors show that across a wide range of assets, the historical volatility process is significantly rougher than the smooth diffusive paths produced by a standard Brownian motion - with estimated Hurst exponents of order $H \approx 0.1$, far below the $H = 1/2$ (BM) assumptions. Motivated by this, practitioners have increasingly replaced the Heston model with a rough counterpart, which drives the variance process using a fractional kernel of exactly the form appearing in the Mandelbrot-Van Ness representation of fBm_H (1.3), rather than a standard Brownian increment.

This substitution comes at a cost since the resulting variance process inherits exactly the pathologies we proved for fBm_H itself in Chapter 1: it is neither a semimartingale (Theorem 1.2.3) nor a Markov process (Theorem 1.2.4). This is a serious practical obstacle, since standard tools for simulation and pricing - Markov-chain-based numerical schemes, Feynman-Kac arguments - rely fundamentally on the Markov property.

We resolve this using the Muravlev representation (1.4.4) developed in Chapter 1, together with its finite-dimensional approximation. This allows us to replace the fractional kernel with a finite superposition of Ornstein-Uhlenbeck kernels, yielding the so-called lifted or Markovian approximation of the rough Heston model - a model which is genuinely Markovian and hence tractable, while still capturing the empirically observed roughness of volatility.

2.1 Black-Scholes & Implied Volatility

A key assumption underlying most derivatives pricing is that, under an appropriate risk-neutral measure, the discounted price process of a traded asset is a martingale. With this in mind, the log-price of an asset is typically modelled as a continuous semimartingale, and the asset price itself as the solution to an SDE of the form

$$dS_t = \mu_t S_t dt + \sigma_t S_t dW_t,$$

where μ_t is the drift, σ_t the volatility process, and W_t a standard Brownian motion. The choice of volatility process σ_t is what distinguishes the models we will consider, and it is central to how well a model's prices match those observed in the market.

The simplest choice, taken by Black and Scholes [2], is to fix $\sigma_t \equiv \sigma$ constant, giving the classical geometric Brownian motion model

$$dS_t = rS_t dt + \sigma S_t dW_t, \quad S_0 > 0,$$

with r the (constant) risk-free rate. This yields the closed-form Black-Scholes price for a European call,

$$C^{BS}(\sigma) = S_t \mathcal{N}(d_+) - K e^{-r(T-t)} \mathcal{N}(d_-), \quad d_{\pm} = \frac{\ln(S_t/K) + \left(r \pm \frac{\sigma^2}{2}\right)(T-t)}{\sigma\sqrt{T-t}}.$$

Since C^{BS} is strictly increasing in σ , any observed market price $C(K, T)$ for a European option with strike K and maturity T has a unique implied volatility $\sigma^{BS}(K, T)$ solving $C(K, T) = C^{BS}(K, T, \sigma^{BS}(K, T))$ - the volatility that, plugged into the Black-Scholes formula, reproduces the observed price exactly. Computing $\sigma^{BS}(K, T)$ across a grid of liquidly traded strikes and maturities produces the implied volatility surface, which is the standard target against which volatility models are calibrated: a model is considered consistent with the market if the implied volatility surface it generates matches the one observed empirically.

If the Black-Scholes model's constant- σ assumption held in practice, this surface would be flat. It is not: real implied volatility surfaces exhibit both a smile (implied volatility rising for strikes far from the money) and a skew (asymmetry between low and high strikes), varying with both strike and maturity. Since Black-Scholes cannot generate any of this variation, it systematically misprices options away from at-the-money, motivating the stochastic volatility models - beginning with the Heston model of Section 2.2 - that we develop for the remainder of this chapter.

2.2 The Heston Model

So clearly the Black-Scholes assumptions of constant volatility are highly unrealistic¹. Thus people nowadays instead model the volatility as a stochastic

¹This would imply that volatility surfaces would be flat, which is not the case empirically.

process too. One of the most realistic of these stochastic volatility models is the Heston model, which is the key model we will be working with throughout this project, and has the following dynamics:

$$\begin{cases} dS_t = S_t \sqrt{v_t} dW_t \\ dv_t = \lambda(\theta - v_t) dt + \sigma \sqrt{v_t} dB_t \end{cases} \quad (2.1)$$

Here the initial value of the underlying asset S_t can be observed from the market while the parameters $\lambda, \theta, \sigma, v_t$ are positive and represent respectively the mean reversion rate, the long run variance, the volatility of the variance and the value of the variance process. Moreover we assume that the risk-free rate $r = 0$ and that the Brownian motions have a correlation factor $\rho = \langle W, B \rangle$.

This model also makes sense from a heuristic standpoint. First of all the asset process agrees with the empirical observation that over small increments the stock price changes proportionally to the square root of the variance. Moreover under this model we have that the assumption that the volatility of variance σ is constant, notice that the variance process is looks very similar to the OU process from section 1.4.1 with the additional $\sqrt{v_t}$ term which makes it an example of a CIR process². The Heston model is very popular due to the fact that it generates reasonable shapes and dynamics for the implied volatility surface and because there is an explicit formula for the characteristic function of the asset logprice. For this reason it is possible to calibrate the model to European options in a very efficient way and then to use the calibrated model in order to price exotic options such as American or Asian-type performing for example a Monte Carlo simulation.

Theorem 2.2.1 (Characteristic Function of Classical Heston Model). *Consider the Heston model in Equation 2.1, and define $X_t = \log(S_t)$ and consider the natural filtration $\mathcal{F}_t = \sigma(W_s, B_s \mid 0 \leq s \leq t \leq T)$ it holds true that*

$$F(t, z, X_t, v_t) = \mathbb{E}(e^{izX_T} \mid \mathcal{F}_t) = e^{A(t) + B(t)v_t + C(t)X_t}$$

Proof. First of all we apply Itô's Lemma to $X_t = \log(S_t)$, then we get

$$dX_t = -\frac{v_t}{2} dt + \sqrt{v_t} dW_t$$

If we apply the Feynmann-Kac formula for the function $F(t, z, X_t, v_t) = \mathbb{E}(e^{izX_T} \mid \mathcal{F}_t)$ we find the PDE:

$$\begin{cases} \frac{\partial F}{\partial t} - \frac{v}{2} \frac{\partial F}{\partial x} + \lambda(\theta - v) \frac{\partial F}{\partial v} + \frac{v}{2} \frac{\partial^2 F}{\partial x^2} + \frac{\sigma^2 v}{2} \frac{\partial^2 F}{\partial v^2} + \sigma \rho v \frac{\partial^2 F}{\partial x \partial v} = 0 \\ F(x, v, T) = e^{izx} \end{cases}$$

Supposing that $F(t, z, X_t, v_t) = e^{A(t) + B(t)v_t + C(t)X_t}$ and imposing that it is a solution we have:

$$\begin{cases} F \left[(\dot{A} + \dot{B}v + \dot{C}x) - \frac{v}{2}C + \lambda(\theta - v)B + \frac{v}{2}C^2 + \frac{\sigma^2 v}{2}B^2 + \sigma \rho v BC \right] = 0 \\ A(T) = B(T) = 0 \\ C(T) = iz \end{cases}$$

²One additional constraint is that the feller condition: $2\lambda\theta > \sigma^2$ must be satisfied. This ensures that the variance remains positive.

We can simplify F so this is a Riccati differential equation that can be solved analytically and so the theorem is proved $\square$

Moreover, as we said above, in [6], the authors show that for a wide range of assets the historical volatility is much rougher than the one reproduced by a standard Brownian motion. Due to this it has become common to replace the standard Heston model with its rough counterpart using a fractional kernel identical to the one used in the Mandelbrot-Van Ness representation.

2.3 Volatility is Rough

The classical assumption - implicit in both Black-Scholes and the standard Heston model - is that volatility evolves as a diffusion driven by ordinary Brownian motion, and hence has Hölder regularity $\frac{1}{2}$. [6] tested this assumption directly against historical data: examining log-volatility time series across a wide range of assets, they estimated the Hölder exponent of the paths using the scaling behaviour of realised power variations, exploiting exactly the relationship between Hölder regularity and increment scaling established in Theorem 1.2.2. Across every asset studied, the estimated exponent was far below $\frac{1}{2}$ - typically $H \approx 0.1$ - indicating that volatility paths are considerably rougher than Brownian motion, not smoother. This finding, robust across asset classes and time periods, motivates replacing the Brownian-driven variance process of the Heston model with one driven by a fractional kernel of Hurst parameter $H \approx 0.1$, giving the rough Heston model which we will explore next.

2.4 The Rough Heston Model

The rough Heston model (RHM) is build up in such a way that the underlying asset S follows the same dynamics of the classical Heston model and the variance v is defined such that

$$\begin{cases} dS_t = S_t \sqrt{v_t} dW_t, \\ v_t = v_0 + \frac{1}{\Gamma(H + \frac{1}{2})} \int_0^t (t-s)^{H-\frac{1}{2}} \lambda(\theta - v_s) ds \\ \quad + \frac{1}{\Gamma(H + \frac{1}{2})} \int_0^t (t-s)^{H-\frac{1}{2}} \sigma \sqrt{v_s} dB_s \\ \quad = v_0 + I_0^{H+\frac{1}{2}} [\lambda(\theta - v_s) ds + \sigma \sqrt{v_s} dB_s], \end{cases} \quad (2.2)$$

where $H \in (0, \frac{1}{2})$. Notice that the idea here is that we are fractionally integrating in such a way that we have paths that are H -Hölder continuous thereby giving us the rough paths that agree with empirical observations. The fact that the volatility in the rough Heston model is no more a semimartingale and a Markov process seems to be a problem for its tractability. However just like in

the standard Heston case, it was demonstrated in [4] that we can (semi) explicit write down the characteristic function for the rough Heston model, where our solution requires us to solve a related fractional Riccati equation.

Theorem 2.4.1 (Characteristic Function of Rough Heston Model). *Consider the rough Heston model given by 2.2 and define $X_t = \log(S_t)$ then we have*

$$\mathbb{E}(e^{iaX_t}) = \exp\left(\theta\lambda I^1 h(a, t) + v_0 I^{\frac{1}{2}-H} h(a, t)\right)$$

where $h(a, \cdot)$ is the solution of the fractional Riccati equation

$$\begin{cases} D^{H+\frac{1}{2}} h(a, t) = \frac{1}{2}(-a^2 - ia) + (ia\rho\sigma - \lambda)h(a, t) + \frac{\sigma^2}{2}h^2(a, t) \\ I^{\frac{1}{2}-H} h(a, 0) = 0 \end{cases}$$

We choose not to include the long derivation of this result which can be found in [4]. The main idea behind the proof is to show that the model can be written as a limiting sequence of another process known as a Hawks process. We then find the characteristic function of each of the Hawks process' and show that this converges to our characteristic function as required. As we discussed in the standard Heston case, the characteristic is only useful for finding the value of the model at the final time step which is useful for the calibration of the model but it is not possible to use it for pricing exotic options which require Monte Carlo simulation. Nevertheless due to the lack of Markovianity of the volatility process the simulation is difficult, thus in order to simulate the Rough heston paths we instead introduce the lifted Heston model from [7].

2.5 The Lifted Heston Model

2.5.1 Markovian Approximation of Rough Heston

In this section we only consider the case where $H \in (0, \frac{1}{2})$ to both simplify the maths and also since this is what is observed in real volatility markets [6]. Now to fix our non-Markovianity issue presented, we follow exactly the strategy of the Muravlev representation developed in Chapter 1: rather than work with the non-Markovian & singular kernel directly, we approximate it by a finite superposition of exponential (Ornstein-Uhlenbeck type) kernels, each of which is Markovian, and for which the aggregate system remains Markovian as a whole. Following [7], for a fixed $n \in \mathbb{N}$ we define the lifted Heston model by

$$dS_t^n = S_t^n \sqrt{v_t^n} dW_t, \quad S_0^n \geq 0, \quad (2.3)$$

where the lifted variance process v_t^n is built from n underlying factor processes $U^{n,1}, \dots, U^{n,n}$ via

$$v_t^n = g_0^n(t) + \sum_{i=1}^n w_i^n U_t^{n,i}, \quad (2.4)$$

$$dU_t^{n,i} = \left(-x_i^n U_t^{n,i} - \lambda v_t^n \right) dt + \sigma \sqrt{v_t^n} dB_t, \quad U_0^{n,i} = 0, \quad \forall i = 1, \dots, n, \quad (2.5)$$

with deterministic function g_0^n , constants $\lambda, \sigma \in \mathbb{R}^+$, weights and mean-reversion speeds $w_i^n, x_i^n \geq 0$, and correlation $\rho dt = d\langle B, W \rangle_t$, $\rho \in [-1, 1]$.

At first glance, (2.5) does not resemble the Ornstein–Uhlenbeck process of Section 1.4.1: there, each factor Z_t^β was driven by an independent Brownian increment with constant diffusion coefficient, and the factors did not interact. Here, two structural differences appear. First, the diffusion coefficient $\sigma \sqrt{v_t^n}$ is state-dependent rather than constant – exactly the same CIR-type square-root diffusion that distinguishes the classical Heston variance process from a standard OU process (Section 2.2). Second, the drift of each factor $U_t^{n,i}$ contains an additional term $-\lambda v_t^n$, beyond the usual linear mean-reversion $-x_i^n U_t^{n,i}$. Crucially, v_t^n in (2.4) is the weighted sum of all n factors, so every factor’s dynamics in (2.5) depend on the entire system, not on $U_t^{n,i}$ in isolation. The factors $U^{n,1}, \dots, U^{n,n}$ are therefore not independent OrnsteinUhlenbeck processes as in Chapter 1, but a coupled system: each retains the linear, mean-reverting form of an OU process, while being nonlinearly entangled with the others through the shared variance v_t^n . The extra drift term $-\lambda v_t^n$ is precisely the term required so that, once the factors are recombined via the weights w_i^n in (2.4), the combined process v_t^n reproduces the correct mean-reverting behaviour of the rough Heston variance process as a whole.

In [7] the author gives a rigorous proof of existence and uniqueness of (2.3)–(2.5). This approximation of the RHM allows to preserve the quality of the classic Heston model and its rough counterpart, and also to speed up the simulation.

Now by observing³ that $\mathbb{E}(U_t^{n,i}) = \int_0^t -\lambda e^{-x_i^n(t-s)} \mathbb{E}(v_s^n) ds$ and taking the expectation on both sides of 2.4 it is possible to find the relation

$$g_0^n(t) = \mathbb{E}(v_t^n) + \lambda \sum_{i=1}^n w_i^n \int_0^t e^{-x_i^n(t-s)} \mathbb{E}(v_s^n) ds \quad (2.4)$$

From now on, we will restrict our attention⁴ to the case in which the deterministic function $g_0^n(\cdot)$ is of the form:

$$g_0^n(t) = v_0 + \lambda \theta \sum_{i=1}^n w_i^n \int_0^t e^{-x_i^n(t-s)} ds, \quad v_0, \theta \geq 0 \quad (2.5)$$

Notice that if $n = 1, w_1^1 = 1$ and $x_1^1 = 0$ we find the classical Heston model. We also have a closed form equation for the characteristic function of the lifted Heston model:

³this can be seen using Ito’s lemma with $e^{tx}U_t$

⁴We restrict to this form as this is precisely the case that ensures that $n = 1$ recovers the classic Heston model as demonstrated. Hence the lifted model is a generalisation of the classic model that converges to the rough model.

Theorem 2.5.1 (Characteristic Function for Lifted Heston). *Consider the evolution described in the lifted Heston model (2.3) - (2.5) and define $X_t^n = \log(S_t^n)$. Then it holds:*

$$\mathbb{E} \left(e^{iaX_t^n} \mid \mathcal{F}_t \right) = \exp \left(\phi^n(t, T) + iaX_t^n + \sum_{i=1}^n w_i^n \psi^{n,i}(T-t) U_t^{n,i} \right), \quad \forall t \leq T$$

where $(\psi^{n,i})_{1 \leq i \leq n}$ solves the following n -dimensional system of Riccati ODEs:

$$(\psi^{n,i})' = -x_i^n \psi^{n,i} + F \left(ia, \sum_{j=1}^n w_j^n \psi^{n,j} \right), \quad \psi^{n,i}(0) = 0, \quad i = 1, \dots, n$$

with

$$F(ia, v) = \frac{1}{2} (-a^2 - ia) + (\rho \sigma ia - \lambda)v + \frac{\sigma^2}{2} v^2$$

and

$$\phi^n(t, T) = \int_0^{T-t} F \left(ia, \sum_{i=1}^n w_i^n \psi^{n,i}(s) \right) g_0^n(T-s) ds, \quad t \leq T$$

A sketch of this proof is given in Appendix B.

Here the problem seems to be the high number of parameters that must be determined because we have to consider $2n + 5$ parameters: 5 come from $(v_0, \lambda, \theta, \rho, \sigma)$ and the other $2n$ from $(w_i^n, x_i^n)_{i=1, \dots, n}$. Nevertheless, in [1] the author provides a parameterization of $(w_i^n, x_i^n)_{i=1, \dots, n}$ in terms of the Hurst index H of a limiting rough volatility model and one additional parameter r_n and so they reduce to 6 once you have chosen the value of H which usually is fixed at 0.1 .

We recall the main idea given in the Muravlev representation for $H < \frac{1}{2}$ in 1.4.1 and state it again in an alternative form for clarity:

Theorem 2.5.2 (Fractional Kernel for $H \in (0, \frac{1}{2})$). *The fractional kernel appearing in Equation 2.2 admits the representation*

$$K(t) = \frac{t^{H-\frac{1}{2}}}{\Gamma(H+\frac{1}{2})} = \int_0^{+\infty} e^{-xt} \mu(dx), \quad \mu(dx) = \frac{x^{-H-\frac{1}{2}} dx}{\Gamma(H+\frac{1}{2}) \Gamma(\frac{1}{2}-H)} \quad (2.6)$$

We may recall that the proof was given earlier in 1.4.1, where we have merely restated it here in a slightly more visually appealing form to make the following convergence proof in the next section easier to follow. Using the above theorem and setting $v_0 = 0$, it is possible to rewrite our rough Heston variance 2.2 as

$$v_t = \int_0^{+\infty} U_t(x) \mu(dx), \quad x > 0$$

$$U_t(x) = \int_0^t e^{-x(t-s)} (\lambda(\theta - v_s) ds + \sigma \sqrt{v_s} dB_s), \quad \forall x > 0$$

This is important because it allows us to look at the dynamics given by Equation 2.2 as a superposition of infinitely many processes $(U^x)_{x>0}$ that have the same dynamics but different mean reversion $x \in (0, +\infty)$.

2.5.2 Convergence to Rough Heston Model

Just as we expect, the key result here is that our lifted Heston model does in fact converge to our rough Heston model as $n \rightarrow \infty$ in the L^2 norm. First of all, it is important to observe that thanks to Equation 2.5 we have that Equation 2.4 can be rewritten as

$$v_t^n = g_0^n(t) + \int_0^t K^n(t-s) (-\lambda v_s^n ds + \sigma \sqrt{v_s^n} dB_s) \quad (2.11)$$

where

$$K^n(t) = \sum_{i=1}^n w_i^n e^{-x_i^n t}, \quad t \geq 0. \quad (2.7)$$

Then, choosing in an appropriate way the values of $(w_i^n, x_i^n)_{1 \leq i \leq n}$, we can prove the convergence in $L^2(0, T)$ of K^n to the fractional kernel $\bar{K}$.

The convergence proof is very involved and follows some quite intricate analysis. However it is not only the key result of this project, but it also sheds some light onto how the parametrisation of our weight and reversion (w_i^n, x_i^n) are determined, which is why we include the proof. The proof follows closely that given in [1] where we add more heuristic details where necessary. We start by proving the main convergence theorem then apply it to our specific set up:

Theorem 2.5.3 (Kernel Convergence in L^2). *Suppose $\alpha = H + \frac{1}{2} \in (\frac{1}{2}, 1)$ and $(r_n)_{n \geq 1}$ be a sequence of real numbers such that*

$$r_n \downarrow 1 \quad \text{and} \quad n \log(r_n) \rightarrow +\infty \quad \text{as } n \rightarrow +\infty \quad (2.8)$$

and define $(w_i^n, x_i^n)_{1 \leq i \leq n}$ such that

$$w_i^n = \frac{(r_n^{1-\alpha} - 1) r_n^{(\alpha-1)(1+n/2)}}{\Gamma(\alpha)\Gamma(2-\alpha)} r_n^{(1-\alpha)i} \quad (2.9)$$

$$x_i^n = \left(\frac{1-\alpha}{2-\alpha} \right) \frac{r_n^{2-\alpha} - 1}{r_n^{1-\alpha} - 1} r_n^{i-1-n/2} \quad (2.10)$$

Then for all $T > 0$ it holds

$$\|K^n - K\|_{L^2(0, T)} \rightarrow 0 \quad n \rightarrow +\infty$$

where $K(t) = \int_0^{+\infty} e^{-xt} \mu(dx)$ and $\mu(dx) = \frac{x^{-H-\frac{1}{2}} dx}{\Gamma(H+\frac{1}{2})\Gamma(-H+\frac{1}{2})}$.

Proof. Set $\eta_i^n = r_n^{i-n/2}$ for $i = 0, \dots, n$ and observe that it holds by construction $w_i^n = \int_{\eta_{i-1}^n}^{\eta_i^n} \mu(dx)$. Then it is possible to write:

$$\begin{aligned} K^n(t) - K(t) &= \int_0^{\eta_0^n} e^{-xt} \mu(dx) + \sum_{i=1}^n \int_{\eta_{i-1}^n}^{\eta_i^n} (e^{-xt} - e^{-x_i^n t}) \mu(dx) + \int_{\eta_n^n}^{\infty} e^{-xt} \mu(dx) \\ &= I_1 + I_2 + I_3 \end{aligned}$$

Thus we can estimate the L^2 -norm of each integral $(I_j)_{j \in \{1,2,3\}}$. Relying on a second order Taylor expansion, along the lines of the proof of Proposition 7.1 in [3], we get

$$\left| \int_{\eta_{i-1}^n}^{\eta_i^n} (e^{-xt} - e^{-x_i^n t}) \mu(dx) \right| \leq C t^2 r_n^{1/2} (r_n - 1)^2 \int_{\eta_{i-1}^n}^{\eta_i^n} (1 \wedge x^{-1/2}) \mu(dx), \quad t \leq T$$

for all $i = 1, \dots, n$, and with $C > 0$ independent of n, i, t . Thus if we consider the summation over all i we get

$$\|I_2\|_{L_2} \leq C \frac{T^{5/2}}{\sqrt{5}} \sqrt{r_n} (r_n - 1)^2 \int_0^{\infty} (1 \wedge x^{-1/2}) \mu(dx)$$

that gives $\|I_2\|_{L_2} \rightarrow 0$ due to the hypothesis 2.8. Moreover thanks to the same hypothesis on r_n we have:

$$\|I_1\|_{L_2} \leq \int_0^{\eta_0^n} \mu(dx) = \frac{(\eta_0^n)^{1-\alpha}}{\Gamma(\alpha)\Gamma(2-\alpha)} = \frac{(r_n)^{-(1-\alpha)n/2}}{\Gamma(\alpha)\Gamma(2-\alpha)} \rightarrow 0 \quad \text{as } n \rightarrow \infty$$

and

$$\|I_3\|_{L_2} \leq \int_{\eta_n^n}^{\infty} \sqrt{\frac{1 - e^{-2xT}}{2x}} \mu(dx) \leq \frac{(r_n)^{(1/2-\alpha)n/2}}{\Gamma(\alpha)\Gamma(2-\alpha)} \rightarrow 0 \quad \text{as } n \rightarrow \infty$$

and so we have proved the theorem. $\square$

This is an important result because it tells us that in the limit the lifted Heston model converges to the rough version which is the one that we want to approximate.

Theorem 2.5.3 tells us that the approximate kernel K^n converges to the true kernel K . However, this alone is not quite enough to conclude that the lifted variance process v_t^n converges to the true rough Heston variance v_t . The reason is that v_t^n is not a simple function of K^n ; rather, it is the *solution* of a stochastic Volterra equation (2.4) in which K^n appears as an input. So what we really need is a guarantee that small changes to the kernel only produce small changes to the corresponding solution - in other words, that the map from kernel to solution is *stable*. Without such a result, it would in principle be possible for $K^n \rightarrow K$ while the solutions v_t^n behaved very differently from v_t .

This stability result is proved in [7] using a Grönwall-type argument (a standard technique for controlling how far apart two solutions of similar equations

can drift over time); we do not reproduce the argument here; instead we take it as a “black box” and use its conclusion. Combining this stability result with Theorem 2.5.3 gives us what we actually want: $v_t^n \rightarrow v_t$ (in an appropriate sense, uniformly on $[0, T]$ in probability). Since the asset price S_t^n is built directly from v_t^n , this in turn gives $S_t^n \rightarrow S_t$: the lifted Heston asset price process converges to the true rough Heston price process as $n \rightarrow \infty$.

Finally, this convergence of the underlying processes carries over to their characteristic functions: the lifted Heston characteristic function of Theorem 2.5.1 converges to the rough Heston characteristic function of Theorem 2.4.1. This is the key fact that justifies everything we set out to do in this chapter - it tells us that the lifted model, which is tractable and Markovian, can be used as a reliable numerical proxy for the rough Heston model, which is intractable and non-Markovian.

Chapter 3

Conclusion & Future Developments

This project set out to give a self-contained, constructive treatment of fractional Brownian motion suitable for an advanced undergraduate reader, culminating in its application to rough volatility modelling. Starting from fractional calculus and the covariance structure of fBm_H , we developed the Mandelbrot-Van Ness and Volterra kernel representation before arriving at the infinite-dimensional Ornstein-Uhlenbeck (Muravlev) representation, which resolves the central practical obstacle in working with fBm_H : its lack of the Markov property. Building on the geometric quadrature underlying the lifted Heston model of [7], we constructed and validated a finite-dimensional Markovian approximation of fBm_H itself - *lifted fBm_H* - for the sub-diffusive regime $H < \frac{1}{2}$, and extended this discretization to the super-diffusive regime $H > \frac{1}{2}$, a case that, to our knowledge, had not previously been treated with this specific quadrature. In doing so we identified and resolved a genuine failure mode of the naive geometric grid in this regime, replacing it with a log-spaced node scheme, and validated both regimes empirically via Monte Carlo estimation of the covariance structure. Chapter 2 then situated this machinery within its motivating application, connecting the Markovian lifting technique to the lifted Heston model and the broader rough volatility literature.

Several natural extensions were beyond the scope of what could be completed in the time available for this project.

On the simulation side, we would have liked to implement and compare the Davies-Harte circulant embedding method alongside the Cholesky method of Section 1.5.1, giving a fuller picture of the trade-offs between exact simulation schemes for fBm_H . Relatedly, it would have been valuable to apply Hurst parameter estimation techniques directly to historical financial data, to demonstrate first-hand - rather than by citing [6] - that realised volatility exhibits the roughness properties motivating the rough Heston model.

On the theoretical side, several of our results were stated with only a sketch

or an empirical justification rather than a complete proof. In particular, a fully rigorous L^2 convergence proof for lifted fBm $_H$ in the $H > \frac{1}{2}$ regime - analogous to Theorem 2.5.3 for the lifted Heston model, but adapted to the log-spaced quadrature we introduce in Section 1.6 - remains an open task; our validation in this regime is empirical only. In particular, the boundary treatment of the quadrature grid at small β for $H > \frac{1}{2}$ - identified in Section 1.3 as a source of residual error, especially as $H \rightarrow 1$ - warrants further investigation, potentially via an adaptive or non-uniform node placement scheme near $\beta = 0$. More broadly, several theorems throughout this report were presented with sketched or omitted proofs where the full argument, while important, was judged secondary to the main constructive thread of the work; a more complete treatment would fill these in. In this sense, the present work is closer to a guided survey that develops and applies the key ideas than to an exhaustive proof-by-proof treatment.

Finally, we deliberately restricted our attention to fBm $_H$ itself and its Markovian lifting, rather than the stochastic Volterra equations underlying rough Heston more directly. A fuller treatment would develop the theory of stochastic integration against Volterra kernels in its own right, which would in turn better motivate the specific structure of the rough Heston variance process. It would also be natural to extend the discussion to other rough volatility models beyond rough Heston, such as the rough Bergomi model, to give a broader view of how the Markovian lifting technique developed here applies across the rough volatility literature.

Appendix A

Simulating the OU Process

Recall the standard OU process with mean reversion β ,

$$dZ_t^\beta = -\beta Z_t^\beta dt + dB_t,$$

introduced in Section 1.4.1. Since the OU process is Gaussian and Markovian, its transition law is known exactly: for $s < t$,

$$Z_t^\beta \mid Z_s^\beta \sim \mathcal{N}\left(e^{-\beta(t-s)} Z_s^\beta, \frac{1 - e^{-2\beta(t-s)}}{2\beta}\right).$$

This gives an exact (as opposed to Euler-discretized) simulation scheme on any time grid: for a step Δt ,

$$Z_{t+\Delta t}^\beta = e^{-\beta\Delta t} Z_t^\beta + \sqrt{\frac{1 - e^{-2\beta\Delta t}}{2\beta}} W, \quad W \sim \mathcal{N}(0, 1),$$

which introduces no discretization error regardless of the step size Δt , unlike a standard Euler scheme.

For the lifted fBm construction, we require the process to be stationary - recall $Z_t^\beta = \int_{-\infty}^t e^{-\beta(t-u)} dB_u$ - rather than started from $Z_0^\beta = 0$. The stationary marginal is $Z_0^\beta \sim \mathcal{N}\left(0, \frac{1}{2\beta}\right)$, and moreover, since each $Z_t^{\beta_j}$ is driven by the same underlying Brownian motion B , the initial values across different mean-reversion rates are correlated:

$$\text{Cov}\left(Z_0^{\beta_i}, Z_0^{\beta_j}\right) = \int_{-\infty}^0 e^{\beta_i u} e^{\beta_j u} du = \frac{1}{\beta_i + \beta_j}.$$

Writing $\Gamma_{i,j} = \frac{1}{\beta_i + \beta_j}$ for this covariance matrix across the quadrature nodes $\beta_{-n+1}, \dots, \beta_n$, we simulate the correlated stationary initial vector via a Cholesky decomposition $\Gamma = LL^\top$: drawing $\xi \sim \mathcal{N}(0, I)$ and setting $Z_0 = L\xi$ gives Z_0 the correct joint distribution. Each subsequent time step is then simulated independently across j using the exact recursion above, sharing the same driving Brownian increment W across all β_j at each step (since all $Z_t^{\beta_j}$ are driven by the same B_t).

Appendix B

Lifted Heston Characteristic Function

Proof sketch. The argument follows the same Feynman-Kac strategy used in the proof of Theorem 2.2.1 (the classical Heston case), but applied to the $(n+1)$ -dimensional Markov process $(X_t^n, U_t^{n,1}, \dots, U_t^{n,n})$ rather than the two-dimensional pair (X_t, v_t) , since it is this larger vector - and not v_t^n alone - that is Markovian by construction.

Applying Itô's lemma to $X_t^n = \log(S_t^n)$ using (2.3) and (2.5), and then the Feynman-Kac formula to

$$F(t, a, X_t^n, U_t^{n,1}, \dots, U_t^{n,n}) := \mathbb{E} \left[e^{iaX_T^n} \mid \mathcal{F}_t \right],$$

yields a linear PDE in $t, x, u_1, \dots, u_n$ analogous to the scalar case, but now with n first-order terms $\partial F / \partial u_i$ in place of the single term $\partial F / \partial v$. The second-order structure is correspondingly richer: a cross-term $\partial^2 F / \partial x \partial u_i$ appears for each factor - exactly as the single cross-term $\partial^2 F / \partial x \partial v$ arose in the classical case from $\rho = \langle W, B \rangle$, since every $U^{n,i}$ is driven by the same Brownian motion B that drives X_t^n - together with cross-terms $\partial^2 F / \partial u_i \partial u_j$ between every pair of factors, reflecting the coupling of all n factors through the shared variance $v_t^n = g_0^n(t) + \sum_{i=1}^n w_i^n u_i$ appearing in the drift and diffusion coefficients of every factor (see (2.5)).

Proposing the exponential-affine ansatz

$$F = \exp \left(\phi^n(t, T) + iaX_t^n + \sum_{i=1}^n w_i^n \psi^{n,i}(T-t) U_t^{n,i} \right),$$

motivated by the affine structure of (2.4)-(2.5) and by analogy with the ansatz $F = e^{A(t)+B(t)v_t+C(t)X_t}$ used in the scalar Heston case - assigning one coefficient function $\psi^{n,i}(t)$ to each factor in place of the single $B(t)$ - and substituting into the PDE, the coefficient of each $U^{n,i}$ produces one ODE in the system stated

above, while matching the remaining (state-independent) terms produces the integral expression for ϕ^n . The coupling between factors enters only through the common term $F\left(ia, \sum_j w_j^n \psi^{n,j}\right)$ appearing in every equation of the system, which is exactly the source of the nonlinear coupling identified in Section ?? . Terminal conditions $\psi^{n,i}(0) = 0$ and $\phi^n(T, T) = 0$ follow from $F(T, \cdot) = e^{iax}$ as in the scalar case.

We omit the full verification that this ansatz solves the PDE, as it is lengthy but mechanical; the complete derivation, together with existence and uniqueness for the resulting Riccati system, is given in [7] and follows the same programme used to derive the characteristic function of the rough Heston model in [4]. $\square$

Bibliography

- [1] Marco Bertolo. Two simulation schemes for the rough Heston model: a comparison. Laurea magistrale thesis in mathematics, Università degli Studi di Padova, Dipartimento di Matematica "Tullio Levi-Civita", September 2024. URL https://thesis.unipd.it/retrieve/feb1c5be-0c07-4ce4-9fb9-61242bfd22e5/Tesi_da_caricare.pdf. Advisor: Prof.ssa Giorgia Callegaro; Co-advisors: Dott. Andrea Pallavicini, Dott. Roberto Daluiso, Dott. Riccardo Longoni.
- [2] Fischer Black and Myron Scholes. The pricing of options and corporate liabilities. *Journal of Political Economy*, 81(3):637–654, 1973. doi: 10.1086/260062.
- [3] Laure Coutin and Monique Pontier. Approximation of the fractional brownian sheet via ornstein-uhlenbeck sheet. *Bernoulli*, 13(4):1110–1133, 2007. doi: 10.3150/07-BEJ6047. URL <https://projecteuclid.org/euclid.bj/1194980472>.
- [4] Omar El Euch and Mathieu Rosenbaum. The characteristic function of rough Heston models. *Mathematical Finance*, 29(1):3–38, 2019. doi: 10.1111/mafi.12173.
- [5] Lawrence C. Evans. *An Introduction to Stochastic Differential Equations*. American Mathematical Society, Providence, RI, 2013. Lecture notes, University of California, Berkeley.
- [6] Jim Gatheral, Thibault Jaisson, and Mathieu Rosenbaum. Volatility is rough. *Quantitative Finance*, 18(6):933–949, 2018. doi: 10.1080/14697688.2017.1393556.
- [7] Eduardo Abi Jaber. Lifting the heston model. *Quantitative Finance*, 19(12):1995–2013, 2019. doi: 10.1080/14697688.2019.1615113.
- [8] Gregory F. Lawler. *Stochastic Calculus: An Introduction with Applications*. American Mathematical Society, 2006. Chapter 7: Fractional Brownian Motion.

- [9] S. C. Lim. Fractional brownian motion and multifractional brownian motion of riemann-liouville type. *Journal of Physics A: Mathematical and General*, 34(7):1301, 2001. doi: 10.1088/0305-4470/34/7/306.
- [10] S. C. Lim and C. H. Eab. *Fractional Calculus and Fractional Processes with Applications to Financial Economics: Theory and Application*. Academic Press, London, 2016. doi: 10.1016/B978-0-12-804248-9.00001-7.
- [11] Benoit B. Mandelbrot and John W. Van Ness. Fractional brownian motions, fractional noises and applications. *SIAM Review*, 10(4):422–437, 1968. doi: 10.1137/1010093.
- [12] Hiroki Masuda. On multidimensional Ornstein-Uhlenbeck processes driven by a general Lévy process. *Bernoulli*, 10(1):97–120, 2004.
- [13] Ivan Nourdin. *Selected Aspects of Fractional Brownian Motion*. B&SS – Bocconi & Springer Series. Springer, Milan, 2012. doi: 10.1007/978-88-470-2823-4.
- [14] Stefan G. Samko, Anatoly A. Kilbas, and Oleg I. Marichev. *Fractional Integrals and Derivatives: Theory and Applications*. Gordon and Breach Science Publishers, Yverdon, 1993.
- [15] Georgiy Shevchenko. Fractional Brownian motion in a nutshell. *Modern Stochastics: Theory and Applications*, 2(3):278–307, 2015. doi: 10.15559/15-VMSTA39. URL <https://arxiv.org/abs/1406.1956>.
- [16] Luke Colin Thorburn. On integral representations of fractional Brownian motion. MSc thesis, University of Melbourne, 2020. URL <https://lukethorburn.com/files/MScThesis-LukeThorburn.pdf>.
- [17] David Williams. *Probability with Martingales*. Cambridge Mathematical Textbooks. Cambridge University Press, 1991. doi: 10.1017/CBO9780511813658.